

LLM-Enabled Smart Welding in Industry 5.0: Defect Reduction, Energy Efficiency, and Human-Centric Manufacturing

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Abstract—Welding is a safety-critical manufacturing operation in automotive, aerospace, energy, marine, and heavy-engineering industries, where joint integrity, energy efficiency, and human safety are closely coupled. This systematic scoping review integrates welding science, digital twins, energy, robotics, and human factors while separating direct large-language-model (LLM) evidence from conventional industrial AI, transferred evidence, and prospective concepts. A reproducible update searched OpenAlex and Semantic Scholar for publications from 1 January 2000 through 26 August 2026 using ten LLM, vision-language, retrieval-augmented-generation, and foundation-model terms combined with welding, followed by a documented Crossref and publisher-website top-up for indexing-lagged records. The combined search identified 302 records; 123 unique records remained after DOI- and title-level deduplication, 28 candidate reports were assessed, and 21 welding-specific publications were included. Fifteen provide quantitative or comparative evaluation, five describe prototypes or preliminary validation, and one is a focused review. The direct literature supports welding-knowledge question answering, multimodal inspection support, data-sparse equation discovery, LLM-initialized numerical prediction, and high-level robot or process planning, but does not establish safety-certified, continuously deployed, closed-loop LLM control of an industrial welding process. Quantitative claims are therefore retained only with source-level validation and context. A safe architecture is proposed in which high-frequency sensor signals undergo synchronization, preprocessing, feature extraction, and temporal encoding before an LLM receives a typed, uncertainty-aware summary; all recommendations remain subject to standards, digital-twin, deterministic safety, and human-approval gates.

Index Terms—Large Language Models (LLMs); Smart Welding Systems; Industry 5.0; Weld Defect Detection; Welding Process Optimization; Decision Support Systems.

I. Introduction

Welding is central to structural and functional components in automotive, aerospace, energy, marine, and

heavy-engineering applications. Joint properties, reliability, and safety depend on process stability, controlled heat input, and effective defect management. Persistent challenges include quality variation, rework, energy demand, fumes, radiation, and other occupational hazards. Industry 4.0 has changed manufacturing through automation, cyber-physical systems, and data analytics. In welding, this transition introduced robotic cells, real-time sensing, and machine learning (ML) for defect detection and process monitoring. Studies report data-driven defect detection and parameter optimization in gas metal arc welding, friction stir welding, and underwater welding [1]–[4]. These methods, however, are often process-specific, data-intensive, and sensitive to changes in materials or operating conditions. Industry 5.0 broadens the objective from automation alone to human-centered, sustainable, and resilient production. In safety-critical applications, this framing preserves expert oversight and treats automation as support for human judgment [5]–[7]. Welding is a relevant setting because human-robot collaboration can combine repeatable robotic motion with an operator’s contextual knowledge and ability to respond to disturbances [8], [9].

Meanwhile, recent AI breakthroughs addressing knowledge representation and optimization problems at the same time have enabled large language models (LLMs) to successfully integrate diverse data types, including welding specs, experiments, inspections, and operator notes, to enable contextual reasoning and flexible human-machine interaction in Industry 5.0 environments [9], [10].

In such systems, human operators retain supervisory and decision-making authority, while robotic systems perform repetitive tasks that require consistent motion. This division of work can support weld quality, energy

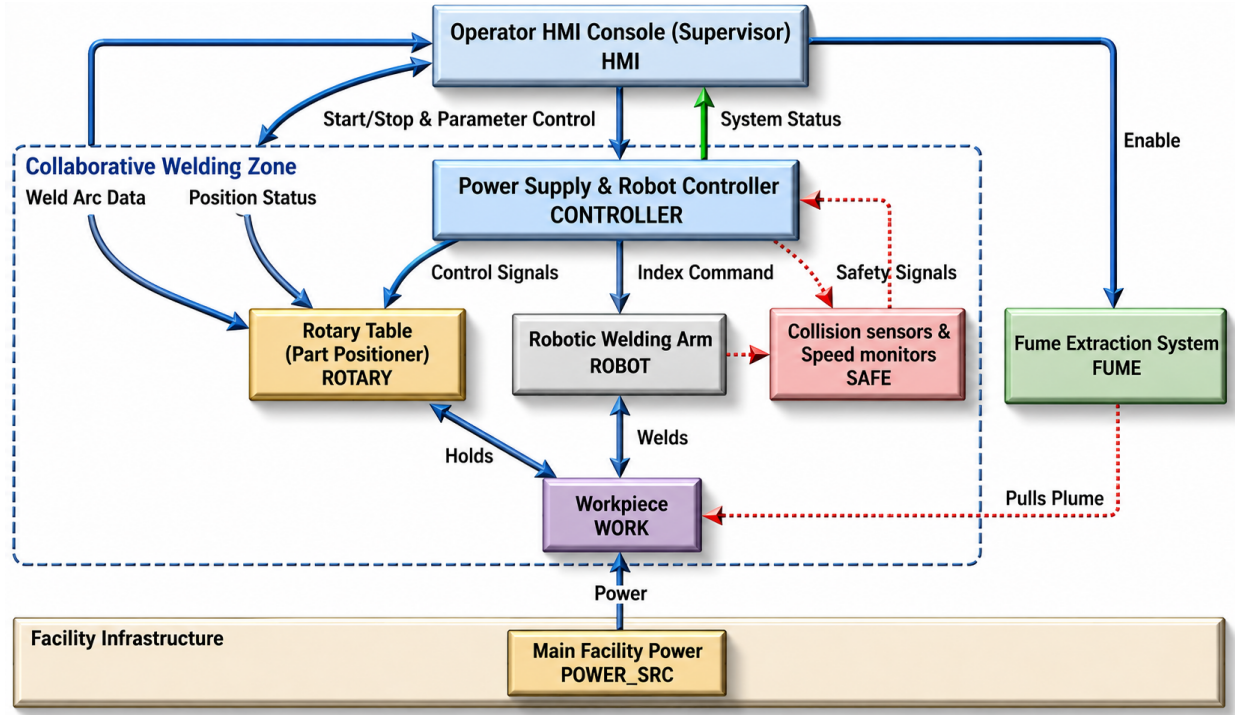


Fig. 1. Illustration of conceptual human-robot collaborative welding cell.

management, and reduced exposure to hazardous conditions, but each benefit requires task-specific validation. The literature on process optimization, defect detection, and ML-based monitoring remains fragmented across particular welding processes and methods. The combined literature on welding science, Industry 5.0 principles, and emerging LLM-enabled knowledge-centric approaches is still in an early stage of development. This review fills that gap by analyzing LLM-enabled smart welding systems during the transition from Industry 4.0 to Industry 5.0. The paper discusses welding processes, welding defects, traditional and data-driven solutions for welding quality and energy optimization, LLM-enabled knowledge management, robotic welding, and human-robot collaboration, with emphasis on evidence limits and future directions for human-centered, resilient, and energy-efficient welding systems.

Three questions organize the evidence: (1) how many publications directly investigate a language-capable LLM, multimodal LLM, vision-language model (VLM), or retrieval-augmented generation (RAG) system for welding; (2) what is experimentally evaluated, transferred from adjacent domains, conventional AI, or prospective; and (3) what sensing and safety architecture is technically credible for human-centered LLM-assisted welding. The updated direct count is 21 publications, but none meets the strict threshold of routine, safety-certified, continuous industrial closed-loop LLM control.

II. Review Methodology

A. Protocol, Information Sources, and Search Date

The review was rebuilt as a reproducible systematic scoping review and reported in a PRISMA-style format [11], [12]. The historical database export used for the first draft was not recoverable as a complete search log; the recovered reference library was used only for backward citation checking and bibliographic reconciliation. No retrospective identification counts were invented. A fresh search was executed in OpenAlex and Semantic Scholar and then updated on 26 August 2026 by a targeted Crossref, publisher-website, and general-web top-up to capture recently registered or indexing-lagged peer-reviewed records. Coverage extends from 1 January 2000 through the final search date.

Ten concept strings were each combined with welding: “large language model,” “large language models,” LLM, “generative pre-trained transformer,” “vision-language model,” “vision language model,” “multimodal large language model,” “retrieval-augmented generation,” “retrieval augmented generation,” and “foundation model.” OpenAlex searched titles and abstracts; Semantic Scholar used bulk scholarly search. The top-up applied the union of these concepts to Crossref and publisher search pages and retained only two additional unique peer-reviewed records after DOI/title comparison with the database corpus: Weld-LLaVA [13] and the WITGS predictor [14]. Full query strings, source-level counts, returned metadata,

TABLE I
Reproducible review protocol

Item	Specification
Databases	OpenAlex and Semantic Scholar; Crossref/publisher web top-up; recovered BibTeX used only for backward checking
Search date	26 August 2026
Period	1 January 2000–26 August 2026
Population	Welding, weld inspection, weld-process knowledge, or welding-robot tasks
Intervention	Language-capable LLM/MLLM/VLM, LLM-based RAG, or LLM-mediated agent
Eligible reports	Peer-reviewed articles/proceedings and focused reviews with a welding-specific method, evaluation, or synthesis
Main exclusions	Conventional CNN/RL/time-series Transformer without language capability; non-welding applications; duplicate preprints or datasets; patents, software records, and non-scholarly notes
Deduplication	DOI first; normalized title when DOI was absent
Quality domains	Model identity, welding directness, data provenance, baselines, validation realism, uncertainty/failure analysis, and reproducibility

deduplication keys, and screening decisions accompany this paper as machine-readable supplementary files.

B. Screening, Eligibility, and Quality Appraisal

Records were screened in two stages. Title/abstract screening required an explicit welding task and a language-capable or language-mediated foundation model. Candidate reports were then assessed against Table I. A CNN containing attention, a Vision Transformer used only as an image classifier, a time-series Transformer, SAM-based segmentation, or an RL policy was not classified as an LLM. Final journal articles superseded matching preprints and repository copies.

Because the included studies range from image benchmarks to RAG systems, equation discovery, simulations, and robot prototypes, a single pooled risk-of-bias score would be misleading. Each report was appraised across seven domains: explicit model identity, welding-task directness, dataset and split provenance, baseline appropriateness, validation realism, uncertainty or failure analysis, and code/prompt/implementation reproducibility. Study-level judgments are supplied in the quality-appraisal file. No pooled accuracy was computed because tasks, labels, datasets, and denominators are not commensurate.

C. Evidence Taxonomy and Transfer Rule

An LLM is a language-token model trained at scale for generation or representation. A multimodal LLM adds encoders or adapters for images or other modalities; a VLM aligns visual and language representations and may or may not be generative. A Transformer is an architectural family, not proof that a model is an LLM. CNNs perform spatial feature extraction, time-series Transformers model numerical sequences, RL learns

TABLE II
Evidence labels used throughout the review

Code	Category	Interpretation
E1	Direct evaluated	Welding-specific LLM/VLM/RAG method with quantitative or comparative evaluation
E2	Direct prototype	Welding-specific architecture, simulation, demonstrator, or preliminary validation
E3	Direct secondary	Review focused on language-capable models in welding
T	Transferred	Adjacent-domain LLM evidence supporting a mechanism, not welding performance
C	Conventional AI	CNN, temporal model, RL, digital twin, or other non-LLM welding evidence
F	Prospective	Proposed capability without direct validation

control policies from reward, and a digital twin is a synchronized virtual representation. These components may cooperate, but their inputs, latency, validation, and failure modes differ.

Contextual work is retained only when a mechanism maps concretely to welding. SteelBERT supports language representations for materials processing but is not a welding trial [15]; CausalKGPT and KG-RAG FMEA support structured industrial reasoning but do not validate weld-defect diagnosis [16], [17]; and human-robot or embodied-robot studies motivate orchestration and safety without demonstrating LLM-controlled welding [18], [19]. Recent manufacturing-wide reviews are used to frame deployment, human oversight, and evaluation gaps, not to inflate the welding evidence count [20], [21]. These sources are labeled T, never E1. Conventional welding ML, DL, RL, sensing, and digital-twin evidence is retained as C because it remains essential background and often supplies the real-time perception or control layer.

III. Evolution of Manufacturing

A. From Craft Production to Cyber-Physical Systems

The history of manufacturing is best understood as a sequence of paradigmatic transformations, each driven by the convergence of new energy sources, information technologies, and production philosophies. These transformations did not occur in isolation; rather, each era built upon the accumulated infrastructure, knowledge, and institutional structures of its predecessor [22], [23].

First Industrial Revolution (1760–1840): The introduction of steam power and mechanized textile production in Britain marked the transition from cottage-based craft production to factory-centered manufacturing. Water- and steam-driven machinery enabled the systematic division of labor, standardization of components, and significant productivity gains. However, production remained largely artisanal in its dependence on the judgment and dexterity of skilled workers.

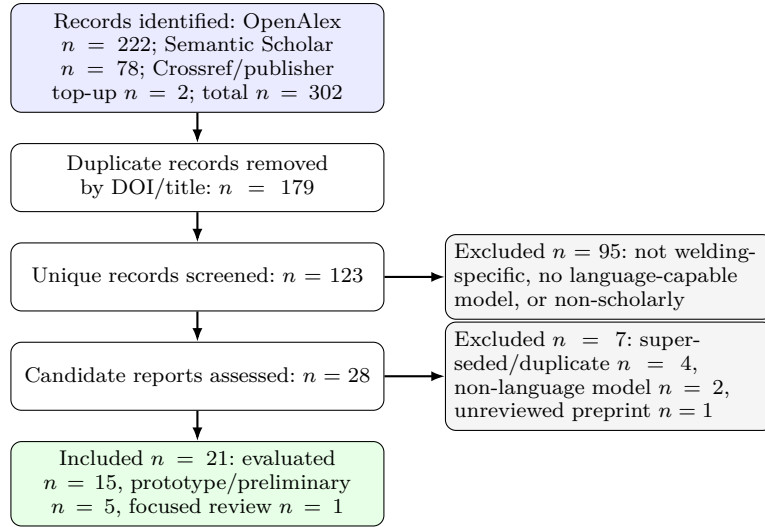


Fig. 2. PRISMA-style flow for the reproducible 2026 update. Counts describe the rerun, not the unrecoverable historical search.

Second Industrial Revolution (1870–1914): Mass production emerged through the application of electrical power, steel manufacturing, and the assembly line. Frederick Winslow Taylor’s scientific management principles introduced systematic time-and-motion studies to optimize workflows, while Henry Ford’s moving assembly line for the Model T demonstrated that standardization and throughput maximization could coexist. This era laid the institutional and organizational foundations for the modern factory [24].

Third Industrial Revolution (1970s–2000s): The proliferation of programmable logic controllers (PLCs), computer numerical control (CNC) machines, and enterprise resource planning (ERP) systems enabled computer-integrated manufacturing (CIM). Production flexibility improved dramatically as reprogrammable automation replaced dedicated hard-wired machinery. Data collection from the shop floor became feasible for the first time, setting the stage for statistical process control and early predictive maintenance frameworks [25].

Industry 4.0 (2010s–present): Cyber-physical systems (CPS), the Industrial Internet of Things (IIoT), cloud computing, data analytics, and advanced robotics increased connectivity among machines, products, and people. In welding, these technologies support robotic cells with multimodal sensing, ML-based defect detection, and digital-twin-assisted process planning [26]–[28].

B. Limitations of the Industry 4.0 Paradigm

Despite its transformative impact, Industry 4.0 has faced growing criticism for its singular focus on automation efficiency and throughput maximization. Several structural limitations have emerged:

- Human marginalization: Automation-centric design often relegated human workers to reactive supervisory

roles, failing to leverage their cognitive flexibility, contextual judgment, and domain expertise [6].

- Data silos and fragmentation: Heterogeneous industrial systems often generated data in incompatible formats, limiting cross-domain reasoning and holistic process optimization [25].
- Brittleness to variability: Rule-based and narrowly trained ML models struggled to generalize across material batches, joint geometries, and environmental disturbances [1].
- Neglect of sustainability: Energy efficiency was treated as a secondary optimization objective rather than a foundational design constraint [7], [24].

These limitations motivated the emergence of Industry 5.0, which explicitly reintroduces the human as a central and irreplaceable actor in intelligent manufacturing ecosystems.

C. The Emergence of Knowledge-Centric and AI-Driven Manufacturing

Parallel to the development of Industry 4.0, a separate but increasingly convergent research trajectory focused on knowledge representation, reasoning, and semantic understanding in manufacturing systems. Early expert systems and ontology-based frameworks attempted to codify engineering knowledge in machine-interpretable forms, but were limited by their rigidity, maintenance overhead, and inability to handle unstructured data.

The emergence of deep learning in the early 2010s and the subsequent development of large-scale pre-trained language and vision models from 2017 onward marked a qualitative shift in AI capabilities. Models such as BERT, GPT, and their successors demonstrated that neural networks trained on massive heterogeneous corpora could develop emergent reasoning, abstraction, and generalization capabilities far exceeding those of task-specific

classifiers [29], [30]. In manufacturing, these capabilities opened new possibilities for cross-domain knowledge integration, natural language human-machine interaction, and adaptive decision support—all of which are central to the Industry 5.0 vision [23].

IV. Role of Machines in Manufacturing

A. Historical Trajectory: From Tools to Autonomous Agents

The role of machines in manufacturing has evolved through several qualitatively distinct phases, each characterized by a different relationship between human agency and machine capability [22], [25].

Passive tools (pre-1800): Machines served as extensions of human physical capability—lathes, forges, and looms required continuous human guidance. Skill resided entirely in the human operator; the machine provided only mechanical leverage.

Mechanized production (1800–1960): Steam and electrically powered machines could execute repetitive operations with greater force and consistency than human workers, but still required human setup, supervision, and adjustment. Quality depended on the knowledge embedded in tooling design and operator expertise [24].

Programmable automation (1960–2000): PLCs and CNC systems allowed sequences of operations to be encoded in software, enabling consistent reproduction of complex tasks without continuous human attention. Machines became repositories of procedural knowledge, encoded in programs by skilled engineers. However, this knowledge was brittle: programs could not adapt to unanticipated variations [26].

Intelligent machines (2000–present): Sensor-rich machines equipped with embedded ML models can now detect anomalies, optimize parameters in real time, and perform condition monitoring autonomously. In welding, robotic systems equipped with vision, acoustic, and electrical sensors have demonstrated the ability to classify defect types and adjust process parameters without human intervention [1], [31].

B. Collaborative Robots and the Shifting Human–Machine Boundary

The development of collaborative robots (cobots) represents a fundamental architectural change in how machines are integrated into manufacturing environments. Unlike traditional industrial robots—which operated behind safety barriers in strictly delimited workspaces—cobots are engineered for physical proximity to humans and require application-specific sensing, interfaces, and safety measures [18].

In welding, cobots have enabled new operational configurations in which humans and machines share workspaces, with task allocation determined dynamically based on complexity, hazard level, and operator workload. Human

welders handle tasks requiring contextual judgment—joint fit-up assessment, material identification, and quality arbitration—while cobots execute repetitive arc and spot welding operations with high positional accuracy [9], [18].

This reconfiguration reflects a broader conceptual shift: machines are no longer conceived primarily as replacements for human labor, but as cognitive and physical partners that amplify human capability while absorbing physical risk and variability [8], [32].

C. Digital Machines: Digital Twins and Cyber-Physical Integration

A defining feature of contemporary intelligent manufacturing is the emergence of the digital twin (DT)—a real-time, bidirectionally coupled virtual replica of a physical asset or process. DTs integrate physics-based simulation, machine learning, and live sensor data to enable predictive control, “what-if” scenario analysis, and lifecycle optimization without physical experimentation [27], [28].

In welding applications, digital twins have been demonstrated for weld pool simulation, thermal field prediction, distortion forecasting, and defect risk estimation [33]. By providing a validated virtual environment in which parameter changes can be evaluated prior to physical execution, DTs significantly reduce the cost of process development and the risk of defect-inducing parameter excursions [33].

From a systems perspective, the DT represents the fusion of the physical and informational roles of machines: it is simultaneously a model of the machine’s behavior and an active participant in its control. This duality—physical execution coupled with virtual cognition—defines the cyber-physical machine of Industry 4.0 and sets the stage for the knowledge-augmented, LLM-enabled machines envisioned in Industry 5.0 [33], [34].

D. Knowledge Machines: LLMs as Manufacturing Agents

The most recent development in the role of machines in manufacturing is the emergence of LLM-enabled agents capable of processing unstructured knowledge—technical standards, inspection reports, operator observations, and scientific literature—and transforming it into actionable decision support [23], [29].

Unlike earlier AI systems that required carefully labeled numerical datasets and task-specific architectures, LLMs can engage in multi-step reasoning across heterogeneous information types, generate explanations accessible to non-expert operators, and dynamically adapt their behavior based on contextual cues. In manufacturing, these capabilities translate into systems that can diagnose defect root causes from maintenance logs, recommend parameter adjustments grounded in standards, and interact with engineers through natural language interfaces [16], [35].

This evolution—from passive tools to programmable machines, from robotic cells to collaborative cobots, from digital twins to LLM-enabled cognitive agents—defines

the trajectory along which manufacturing intelligence has developed and provides the technological and conceptual foundation for the Industry 5.0 paradigm discussed in the following section.

V. Industry 5.0 and Human-Centric Manufacturing

A. Definitions and Vision

Industry 5.0 was conceived as a corrective and complementary approach to Industry 4.0. Industry 5.0 redirects the emphasis of manufacturing systems from technology-based automation towards human-centric, sustainable, and resilient manufacturing systems. While Industry 4.0 relies on cyber physical systems, data integration, and self-sovereign decision-making, Industry 5.0 reinstates the human actor as a key player whose cognitive capacities, creativity, and contextual judgment are seen as crucial elements in controlling complex industrial environments [9], [32].

The key vision of Industry 5.0: human machine collaboration in a synergistic manner, where intelligent machines are conceived to collaborate with humans instead of replacing them. This paradigm explicitly considers personalization, enabling manufacturing systems to accommodate individual worker capacities and specific customer requirements while maintaining high efficiency and quality standards [7].

While Industry 4.0 primarily considers optimizing throughput and automation, Industry 5.0 considers meaningful human participation, ergonomic integration, and transparency of decisions within manufacturing processes [6].

Sustainability represents a second key foundation for Industry 5.0, going well beyond the conventional notion of “energy sustainability.” Rather, it also includes social sustainability and long-term environmental sustainability. From an empirical analysis, it was found that the adoption of Industry 5.0 is related to social sustainability, rather than technological, as it relates to worker safety, skill building, and organizational adaptability, rather than technological readiness itself [7].

The change in focus represents a broader vision for industry, where productivity is balanced against environmental and social limitations, rather than Industry 4.0, which remains automation-centric, with humans in the loop as system supervisors rather than active participants. As a result, studies on the review of design in the manufacturing system reveal that automation-centric approaches to Industry 4.0 design face challenges in dealing with variability, material uncertainty, and human-dependent decision processes, which motivates the move to Industry 5.0 design principles [6].

B. Human–Robot Collaboration (HRC) in Industry 5.0

Human Robot Collaboration (HRC) is one of the cornerstones of Industry 5.0, and this is a fundamental change from automated manufacturing to cooperation-oriented manufacturing processes. A key aspect of this change is represented by collaborative robots, or “cobots,” which are designed to cooperate with humans in a safe and intuitive way. Unlike traditional industrial robots, which are typically installed in a closed and highly controlled environment, collaborative robots are sensitive to human behaviors and are able to adapt in real time to circumstances, thus allowing humans and machines to share their working space and to cooperate in a seamless way, rather than working independently from one another [9].

Instead of the human operators being seen as interchangeable parts of the process within the automated workflows, Industry 5.0 repositions the human operators as a key part of a cognitive collaboration with the robots. The focus is no longer on achieving the maximum output from the process but rather on establishing a cognitive partnership with the robots in which tasks are allocated according to the strengths of each entity. Physically demanding tasks, repetitive tasks, and precision-intensive tasks are left to the robots, whereas the human operators retain the cognitive tasks that involve judgment, creativity, and adaptation. The empirical evidence indicates that workflows that are designed around the strengths of the human operators not only improve usability and effectiveness without compromising productivity but also reduce physical and cognitive fatigue [32].

One of the key advancements in Industry 5.0-enabled HRC is the development of dynamic and human-aware scheduling systems. Building upon the static scheduling systems of Industry 4.0 and earlier, Industry 5.0’s dynamic systems now consider human energy and cognitive levels to be key variables for the operation of a production system. With the ability to integrate real-time data on human fatigue and cognitive levels into the scheduling process, it is now possible for production systems to autonomously adjust the allocation of tasks between humans and robots. This not only supports the environmental sustainability objectives of Industry 5.0 but also maintains the well-being of the workforce by ensuring that production targets are achieved without compromising human performance [8].

This is a clear move away from the static scheduling systems of Industry 4.0, which failed to account for the variability in human performance and assumed the same level of performance from operators. Decision-support interfaces can translate complex robot and process states

into information that operators can inspect and challenge. Such interfaces may support situational awareness, but trust and resilience depend on explanation quality, calibrated uncertainty, training, and evidence from the intended workplace [9].

C. Sustainability and Energy Efficiency in Industry 5.0

Industry 5.0 significantly transforms the conventional understanding of sustainability in manufacturing from a more narrow energy efficiency perspective to a more comprehensive one based on three integrated pillars of environmental responsibility, social responsibility, and system robustness. Sustainability in Industry 5.0 differs from Industry 4.0, in which sustainability was often viewed as a secondary optimization objective in manufacturing. In Industry 5.0, sustainability is an integral part of manufacturing systems. This approach enables a manufacturing environment in which efficiency, sustainability, and responsibility for the environment and employees are balanced, and sustainability becomes an integral part of manufacturing systems rather than an external factor. This provides a more comprehensive understanding of sustainability in manufacturing [7].

Empirical studies on manufacturing processes reveal that an increase in energy consumption, despite enhancing manufacturing efficiency in the short term, may compromise manufacturing efficiency in the long term. Similarly, studies on advanced manufacturing processes reveal that variability in manufacturing processes may increase rework and inefficiencies in resource consumption and energy consumption. This reveals a limitation of energy-intensive optimization approaches and emphasizes the need for predictive approaches based on knowledge, which are more relevant to Industry 5.0. [36].

Advanced planning methodologies, which may be enabled via large language models, provide another opportunity to minimize energy waste via a combination of historic process knowledge and context-dependent operational suggestions. A key aspect of such intelligent technologies is that they are intended to augment, rather than replace, human decision-makers. This allows them to more effectively assess sustainability-driven trade-off possibilities within complex manufacturing environments. This form of human-driven optimization aligns well with the fundamental Industry 5.0 philosophy of balancing technological intelligence with human judgment to produce more sustainable manufacturing possibilities as shown in Table I [9].

TABLE III
Comparison of Industry 4.0 and Industry 5.0 paradigms

Aspect	Industry 4.0	Industry 5.0
Core focus	Automation and digitalization	Human-centric intelligence
Role of humans	Supervisors of automated systems	Active collaborators and decision-makers
System objective	Efficiency and throughput	Sustainability, resilience, and personalization
Human-machine interaction	Limited, interface-driven	Ergonomic, adaptive collaboration
Sustainability	Secondary optimization goal	Embedded design principle
Decision support	Algorithm-centric	Human-AI co-decision frameworks

TABLE IV: Contextual evidence map for LLM-enabled manufacturing and welding

Ref	Process	AI/Method	Objective	Human Role	Limitation
This table preserves the manuscript’s broad interdisciplinary context. Rows outside welding are transferred evidence (T); welding studies without a language-capable model are conventional AI (C). Only the separately screened publications in Table VII contribute to the direct LLM-welding corpus (E1–E3).					
[23]	Manufacturing	AI Survey	State-of-the-art review	Strategic	No unified benchmarks
[24]	Manufacturing	Thermodynamic	Energy analysis	Interpretive	Human factors ignored
[24]	Manufacturing	Resource Analysis	Efficiency study	Interpretive	Static modeling
[25]	Smart Mfg	Data-driven	Predictive manufacturing	Supervisory	Data silos
[26]	Manufacturing	Cloud-CPS	Resource sharing	Strategic	Latency concerns
[37]	Industrial IoT	Survey	IoT architectures	Supervisory	Security challenges
[27]	Smart Mfg	Digital Twins	Closed-loop optimization	Supervisory	Integration complexity
[22]	Manufacturing	Industry 4.0	Intelligent mfg vision	Strategic	Human-centric gap
[38]	Digital Eng.	Auto-config	Low-cost SME solutions	Decision support	Limited autonomy
[6]	Mfg Systems	TMPE Framework	Review design enablers	Strategic	Limited guidance
[34]	Edge-Cloud	Hetero Nodes	Uncertainty handling	Supervisory	Network dependency
[29]	Logistics	Found. Models	Digital engineering	Strategic	Indirect mfg focus
[39]	Built World	AI Survey	Digital twins	Strategic	Domain specific
[33]	Manufacturing	Digital Twins	Closed-loop monitoring	Supervisory	Model complexity
[28]	Product Life.	Digital Twins	Design–mfg integration	Supervisory	Interoperability
[40]	Robotics	3D Recon.	Environment perception	Supervisory	Deployment immaturity
[41]	Additive Mfg	Introduction	Process fundamentals	Interpretive	Not data-driven
[42]	EV Recycling	Framework	element extraction	Strategic	Immature system
[43]	Battery Mfg	AI + Sensing	Fault diagnosis	Supervisory	Scalability
[44]	Giga-casting	AI Framework	Die design optimization	Supervisory	High validation cost
[33]	Robotics	Digital Twins	Dynamic adaptation	Supervisory	Computational cost
[33]	Mfg Systems	Digital Twins	Intelligent systems	Supervisory	Scalability challenges
[45]	Policy	Conceptual	Industry 5.0 vision	Central	Lacks roadmap
[45]	Manufacturing	Conceptual	Human-centered transition	Central	No system metrics
[6]	Industry 5.0	Survey	Review enabling tech	Strategic	Conceptual overlap
[9]	HMI	Adaptive Int.	Decision support	Central	Cognitive overload
[32]	IMS	Proactive Design	Human-centric usability	Co-designer	Scaling issues
[7]	Manufacturing	Empirical Study	Algeria case study	Strategic	No AI validation
[8]	Job Shop	Optimization	Human-aware scheduling	Decision-in-loop	Limited real-time use
[5]	DFM	Generative DL	Proactive design	Co-designer	Subjectivity
[46]	Ergonomics	Biomechanical	Workload reduction	Assisted	Limited AI focus
[8]	Ergonomics	Deep Learning	Workload estimation	Central	Sensor intrusive
[18]	HRC	AI-enhanced	Safe collaboration	Collaborative	Pre-LLM focus

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TABLE IV – Continued from previous page

Ref	Process	AI/Method	Objective	Human Role	Limitation
[47]	HRC	Review	Classify HRC models	Collaborative	Pre-Industry 5.0
[18]	Robotics	Survey	Industrial HRC	Collaborative	Limited autonomy
[18]	HRC	Multimodal ML	Predict human intent	Collaborative	Noise sensitivity
[48]	Smart Mfg	XR + AI	Enhanced HRC	Collaborative	Hardware dependence
[46]	Ind. Tasks	Supervised ML	Skill quantification	Performer	Task-specific
[48]	Manufacturing	Data-driven ML	Assess human skill	Performer	Sensor dependent
[23]	Manufacturing	AI + KM	Decision support	Decision-in-loop	Explainability gap
[49]	Welding Physics	Physical Mod.	Theory of welding	Interpretive	No automation
[50]	Metallurgy	Theory	Microstructures	Interpretive	No predictive AI
[51]	Laser Welding	Synchrotron	Keyhole dynamics	Observational	No control
[52]	X-ray Inspect.	Review	Defect recognition	Supervisory	Dataset scarcity
[3]	Friction Stir	Review	FSW fundamentals	Interpretive	No AI integration
[50]	Electron Beam	Review	Aerospace apps	Interpretive	Manual inspection
[50]	Fusion Welding	Review	Cracking prevention	Interpretive	No predictive models
[53]	Welding Tech	Review	Engineering apps	Supervisory	Qualitative only
[54]	Pressure Vessel	Ultrasonic	Detection capability	Interpretive	Manual operation
[55]	Reactor Piping	Fracture Mech.	Seismic reliability	Interpretive	No automation
[56]	Materials	Continuum	Lattice defects	Interpretive	Theoretical only
[57]	Boilers	SEM/AES/SIMS	Corrosion damage	Interpretive	No AI prediction
[58]	Friction Elem.	Control Charts	Joint monitoring	Supervisory	Statistical only
[36]	Laser Welding	Review	AHSS forming	Interpretive	No AI control
[1]	Weld Pool	Sensing	Dynamics analysis	Observational	No AI autonomy
[10]	Friction Stir	ML Regression	Quality prediction	Supervisory	Generalization limit
[2]	Laser Welding	Random Forest	Porosity monitoring	Monitoring	Sensing cost
[1]	Welding	ML/DL Survey	Decade review	Supervisory	Data scarcity
[59]	Surface Images	Neuro-Fuzzy	Defect classification	Supervisory	Complex tuning
[60]	Resistance Spot	Ensemble DL	Multimodal fusion	Supervisory	Interpretability
[61]	Busbar Weld	ML Models	Spectrum/Vision fusion	Supervisory	Dataset bias
[31]	GMAW	Deep Learning	Online quality prediction	Supervisory	Noise sensitivity
[62]	High-Res Weld	SCLT Network	Entropy quantization	Decision support	Compute demand
[63]	Weld Detection	CNN Survey	Review architectures	Supervisory	Deployment immature
[64]	Industrial AI	Survey	Defect detection	Supervisory	Unresolved bias
[1]	Welding	ML Survey	Detection techniques	Supervisory	Implementation gap
[65]	Radiographic	CNN	Defect classification	Supervisory	High label effort
[66]	Additive Mfg	ML Review	Bibliometric analysis	Strategic	No operational data
[67]	Defect Detect.	DCNN	Review of models	Supervisory	Hardware limited
[68]	Oil/Gas Pipe	Deep Learning	Intelligent recognition	Supervisory	Label subjectivity

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TABLE IV – Continued from previous page

Ref	Process	AI/Method	Objective	Human Role	Limitation
[69]	X-ray Weld	Multiscale CNN	Adaptive enhancement	Supervisory	High resolution cost
[70]	Laser-MAG	Stacking-PSO	Multi-source fusion	Supervisory	Processing latency
[71]	Battery Weld	Data-driven ML	Strength assessment	Supervisory	Limited to Al-Cu
[72]	X-ray Weld	CNN-Trans.	Multi-scale fusion	Supervisory	Missing small defects
[73]	Laser Welding	Deep RL	Control learning	Supervisory	Training data needs
[74]	Laser Welding	RL	Closed-loop control	Autonomous	Latency issues
[75]	FSW	RL	Parametric optimization	Supervisory	Material specific
[76]	Rail Grinding	RL	multi-obj optimization	Supervisory	Static constraints
[77]	Laser Welding	Neural RL	Intelligent architecture	Supervisory	Initial prototype
[78]	Laser Welding	LLM (T2EGPT)	Derive physical eqns	Human-in-loop	Data curation heavy
[79]	Alloy Design	NLP + LLM	Extract SPP relations	Human-in-loop	Validation needed
[80]	Welding	LLM/DL Review	Precision processing	Supervisory	Domain specificity
[17]	FMEA	Graph + RAG	Risk analysis	Decision-in-loop	Graph overhead
[16]	Aerospace	CausalKGPT	Root-cause analysis	Decision-in-loop	Knowledge drift
[35]	Bridge Steel	LLM + RAG	Causal defect analysis	Decision-in-loop	Interpretability
[81]	Visual Recog.	LLM-centric	Open-set recognition	Supervisory	Efficiency
[15]	Steel Design	SteelBERT	Property prediction	Human-in-loop	High validation cost
[16]	Quality Diag.	Causal LLMs	Fault explanation	Decision-in-loop	Graph cost
[82]	Industrial AI	RAG	Trustworthy systems	Decision-in-loop	Hallucination risk
[83]	Quality Control	Advanced RAG	Ceramic tile diagnosis	Decision support	Domain adaptation
[84]	HRC	LLM Agent	Proactive collab	Collaborative	Autonomy limits
[85]	Additive Mfg	Vision-Lang.	Automated qualification	Supervisory	Annotation burden

VI. Welding Fundamentals

Welding is a fundamental manufacturing process that permanently joins materials using heat, pressure, or a combination of both. Conventional methods of welding processes were mainly based on the expertise of the operator and the selection of parameters through empirical methods, whereas contemporary methods incorporate physics-based simulation techniques along with data-driven and artificial intelligence-based models for predicting the optimal process parameters in advance, thereby reducing the need for experimentation and ensuring high process reliability as shown in Table II [4].

A. Classification of Welding Process

Welding techniques are classified based on the physical bonding and the environmental conditions [4].

B. Overview of Welding Processes and Defect Taxonomy

Figure 3 provides a concise overview of the conventional workflow and taxonomy of industrial joining in one diagram, with processes and topics including: (i) pre-weld, (ii) primary arc and solid state welding, (iii) special/high-energy processes, (iv) post-weld joint evaluation, (v) shared metallurgical issues, and (vi) representative application domains, such as aerospace, automotive, shipbuilding, oil and gas, and robotics. This diagram is meant to be a concise reference guide showing the relationships between each welding family and the defects and inspection methods most frequently encountered with them

1) Pre-weld and Surface Preparation: Good welding starts with good design and preparation of the weld joints to be made. Common pre-weld operations, as depicted in Figure 2, include defining the type of weld required (e.g., butt, lap, tee, corner, edge welds), degreasing and oxide removal, and edge beveling where required to ensure adequate root penetration during welding. Poor pre-weld preparation is often responsible for lack of fusion, porosity, and other weld defects in the final weld joint as shown in Table III [53].

2) Arc Welding Processes and Typical Defects: The central column of the diagram lists conventional arc processes—SMAW (stick), GMAW (MIG/MAG), GTAW (TIG) and FCAW (flux-cored). These processes form the backbone of industrial joining and each has a characteristic defect profile:

- SMAW (Shielded Metal Arc Welding) is widely used for general construction and repair; typical problems include porosity, slag inclusions, lack of fusion and excessive spatter when parameters or electrode handling are not correct [53].
- GMAW (MIG/MAG) is common in automated and semi-automated production. If shielding or parameters are wrong it produces porosity, burn-through, and cold lap defects [53].

TABLE V
Comprehensive Classification of Welding Processes and Their Key Characteristics

Welding Category	Bonding Mechanism	Key Characteristics
Fusion Welding (Arc)	Coalescence via electric arc heating of base and/or filler metal.	Highly versatile; high electrical noise signatures; includes SMAW, TIG, and MIG/MAG.
Fusion Welding (Power Density)	Focused high-intensity laser or electron beams melt the interface.	Deep penetration; narrow Heat Affected Zone (HAZ); high-frequency acoustic emissions.
Solid-State Welding	Bonding below melting point through intense plastic deformation or friction.	Avoids solidification defects; ideal for dissimilar metals (e.g., Al to Cu) and phase-sensitive alloys.
Resistance Welding	Coalescence via heat from electrical resistance under applied pressure.	High-speed automation; no filler material; characterized by rapid thermal cycles.
Hybrid Welding	Synergetic combination of two heat sources (e.g., Laser + Arc).	Improved gap bridging; higher welding speeds; complex multimodal data signatures.
Underwater Welding	Submerged joining in wet environments or hyperbaric dry habitats.	Extremely high cooling rates (10^1 – 10^2 K/s); high risk of hydrogen-induced cracking and porosity.

- GTAW (TIG) yields high-quality welds but is sensitive to tungsten inclusions, crater cracking and oxide (“sugar”) formation on some alloys when shielding or contamination occur [53].
- FCAW (Flux-Cored Arc Welding) is productivity-oriented and can suffer from excessive spatter, undercut, slag entrapment and underbead cracks if flux composition or travel speed is inappropriate [53].

These defect classes (porosity, inclusions, undercut, lack of fusion, spatter, crack modes) are the primary quality concerns for fusion welding and drive both process selection and inspection strategy.

3) Special / High-Energy Processes: The figure also highlights high-energy processes—laser beam welding, electron beam welding and resistance (spot) welding—which are widely used where high precision, low heat input or very fast cycle times are required:

- Laser beam welding enables high travel speeds and narrow fusion zones but can produce keyholing-related defects, underfill, and hump formation when power/velocity balance is incorrect [86].

Recent reviews stress that each high-energy process has a distinct set of process windows and signature defects, so the choice between them is driven by part geometry, materials and acceptable defect types [50], [86].

4) Joint Evaluation: NDT and Destructive Testing: The final column of the figure groups joint evaluation techniques into non-destructive testing (NDT) and destructive testing (DT) methods. Standard NDT methods—visual testing (VT), liquid/ dye penetrant testing (PT), magnetic particle testing (MT), ultrasonic testing (UT) and radiography (RT)—provide complementary detection capabilities for surface and subsurface defects; modern surveys emphasise that combining multiple NDT modalities and advanced signal processing (including ML) improves detection and reduces false positives [52]. Destructive testing (tensile, bend, impact, hardness mapping, nick-break) remains essential for qualification and mechanical property verification of welded joints.

5) Shared Metallurgical Challenges: The figure groups several metallurgical failure modes that recur across welding technologies:

- Hot cracking (solidification cracking) appears during solidification of the weld metal and is strongly dependent on alloy composition and solidification path; solidification crack tests and susceptibility criteria are active research topics [50].
- Cold cracking, often hydrogen-induced, frequently occurs in the heat-affected zone (HAZ) after solidification when residual tensile stresses and diffusible hydrogen combine [53].
- Metallurgical degradation (excessive HAZ grain growth, loss of corrosion resistance, embrittlement) is another cross-cutting concern that sets filler metal, preheat and interpass temperature requirements [50], [53].

6) Digital Twins, Real-time QA and Applications: Modern welding research increasingly combines sensing, digital twins and AI to enable in-process monitoring and weld pool prediction: digital twin approaches reconstruct weld pool evolution and enable predictive control to reduce defects, bridging the gap between process physics and real-time quality assurance [33]. Typical application domains shown in the figure—aircraft fuselage and engines, automotive chassis and exhausts, ship hulls, structural steel, pipelines and robotic fabrication—reflect industries where reliability, throughput and inspection cost tradeoffs motivate choice of process and inspection strategy [33], [53].

7) Figure Interpretation — how to read the map: The left column groups the high-level workflow (pre-weld → processes → evaluation → challenges → applications). Each process box points to typical defect types (colored boxes in the figure) and to the inspection and testing methods most appropriate for detecting the listed faults. Use the map as a rapid checklist: for example, if a part is to be joined using GMAW, anticipate porosity and spatter concerns and plan appropriate NDT (VT/UT/RT) and process control; if laser welding is selected for a thin aerospace skin, account for keyhole instability and integrate inline sensing or a digital twin to control the keyhole regime [33], [52], [86].

8) Microstructural Zones in FSW: Friction stir welding (FSW) generates a different microstructural regions across the weld because the material experiences uneven heating and deformation during the process. Based on the severity of thermal and mechanical effects, the FSW joint conventionally divided into four zones.

C. Evolution of Welding Technology Toward Smart Welding

Figure 3 shows the evolution of welding technology from conventional manual welding to advanced intelligent and human-centric manufacturing technologies. As shown in the figure, there are five major eras in the evolution of welding technology: the Traditional Era, Automation Era, Digital Era, Intelligence Era, and the current Smart Welding Era.

1) Traditional Era (1900s–1960s): The first stage in the industrial welding evolution is manual welding carried out by expert welders using various welding processes, including Shielded Metal Arc Welding (SMAW) and early stages of Gas Metal Arc Welding (GMAW). During this stage, the quality of welds depended mainly on the expertise and manual skills of the welders. Human factors played an important role in controlling various welding parameters, including arc length, travel speed, and electrode angle. As a result, welding processes and weld quality were subject to variability, including inconsistent quality. The quality of welds was not consistent due to the

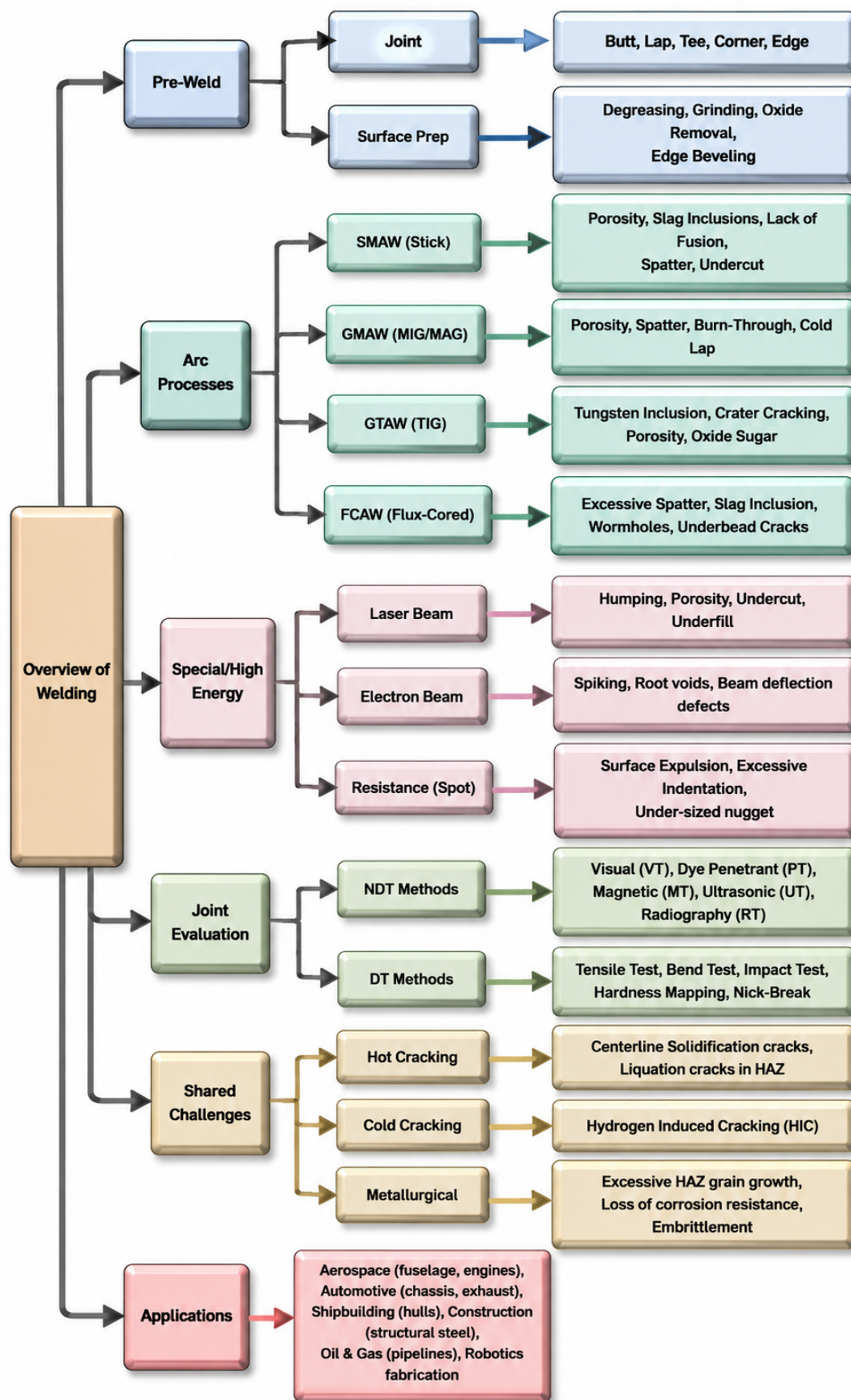


Fig. 3. Overview of welding processes, common defects, joint evaluation methods

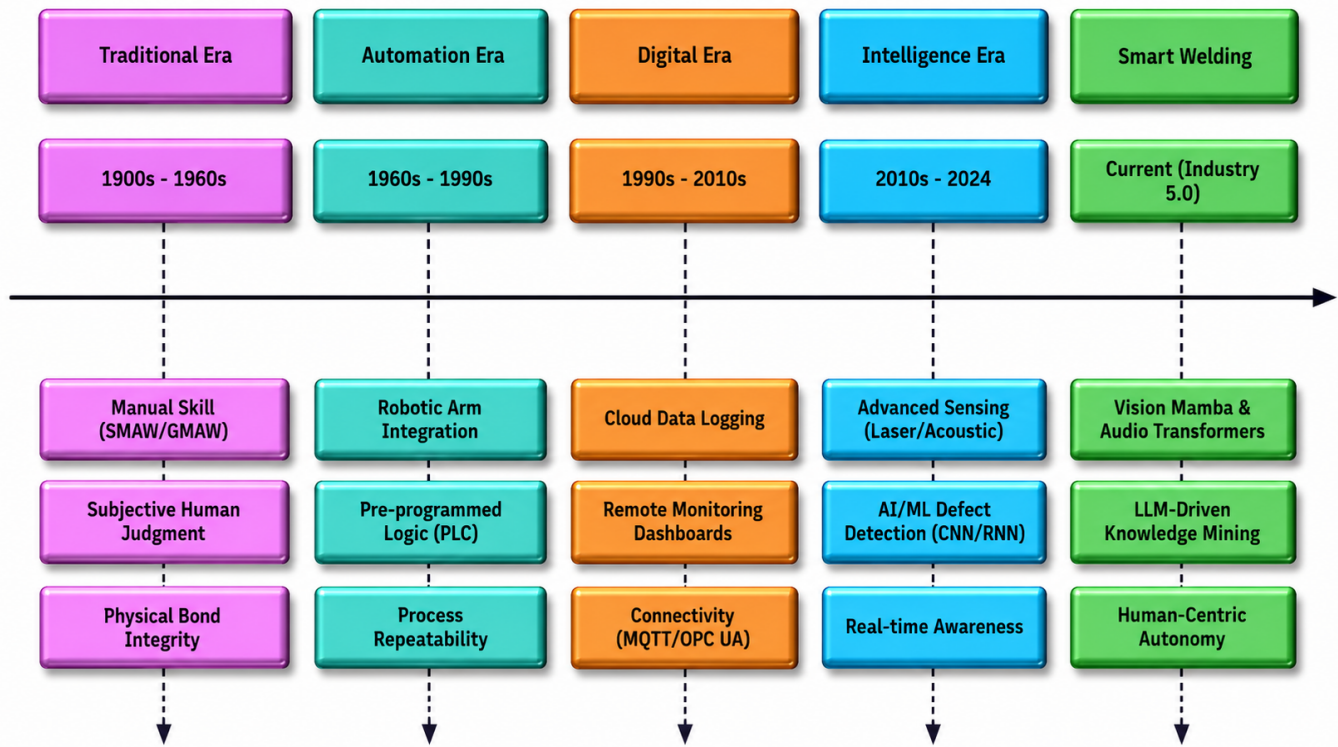


Fig. 4. Evolution of Welding

manual skills and expertise of the welders. Inspection was mostly visual or mechanical in nature, and defect detection was heavily dependent upon human expert knowledge. Welding quality assessment was mostly concerned with ensuring physical bond integrity and reliability rather than optimizing welding processes. There was a lack of automated sensing and monitoring, and so there was no way to detect defects during the actual welding process. With an increase in industrial production, this was a problem that needed to be overcome.

2) Automation Era (1960s–1990s): This stage corresponds to the implementation of automation in welding processes. Automation technology was incorporated in welding processes with the introduction of industrial robots, especially in automotive manufacturing, where repetitive welding tasks had to be performed. The robotic arms facilitated precise control of welding torch movements and ensured better reproducibility of process parameters. In this era, programmable logic controllers (PLCs) were also implemented to control welding processes. PLC-based welding processes allowed engineers to program welding processes and maintain consistent process parameters during high-volume production processes. This greatly reduced human error and increased productivity. The robotic welding systems also allowed for better thermal management and consistency in the quality of the weld beads for multiple parts. However, the early robotic

systems were mainly capable of functioning through pre-programmed paths and were not adaptable. They were not capable of responding to changes in the geometry of the joints, material thickness, or environmental factors. Despite these limitations, the introduction of the robotic welding systems marked a major breakthrough in the history of welding system automation and paved the way for the development of modern cyber-physical production systems.

3) Digital Era (1990s–2010s): The digital era saw the introduction of data acquisition, connectivity, and remote monitoring capabilities in welding systems. Advances in computer and industrial network technologies enabled welding systems and robots to produce large quantities of data regarding the welding process, such as current, voltage, speed, temperature, and arc signals.

Cloud-based data logging and remote monitoring dashboards became a common feature in industrial processes. MQTT and OPC-UA communication protocols enable the welding machines to send real-time operation data to centralized monitoring platforms. This technology allowed engineers to analyze welding parameters in entire industrial processes and detect abnormal welding conditions. The digitalization of welding processes also allowed for predictive maintenance and optimization strategies to be implemented in welding processes. Historical welding process data was analyzed to detect patterns associated with

welding defects or equipment degradation. This marked a transition from mechanical to data-driven manufacturing processes and is a critical foundation for Industry 4.0 manufacturing architectures.

4) Intelligence Era (2010s–2024): This phase marked the addition of advanced sensing and artificial intelligence to welding processes. Advanced sensing devices such as laser scanners, acoustic sensors, infrared cameras, and high-speed cameras are now a common feature in monitoring weld pool and geometry in real time. Artificial intelligence and machine learning algorithms have been utilized to a greater extent for weld defect detection, process optimization, and weld seam tracking. Deep learning architectures such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and transformers have proven to analyze multi-modal sensor data to detect weld defects such as porosity, lack of fusion, and undercut during the welding process. Such intelligent monitoring systems help to attain real-time awareness of welding conditions, and automated corrective actions can be performed accordingly. For instance, an adaptive welding controller can be designed to adjust the welding current or travel speed based on abnormal behaviors of the weld pool during welding. Studies have proven that incorporating intelligent monitoring systems enhances welding quality and reduces production defects by detecting abnormalities in welding processes early on [87].

5) Smart Welding and Industry 5.0: The final step in this evolution process described in the figure corresponds to the emerging concept of smart welding as a part of Industry 5.0. Indeed, smart welding systems utilize advanced sensors, artificial intelligence, digital twin technology, and human machine collaboration in order to create a highly adaptive manufacturing environment. The recent advances in this field include the use of digital twin technology in order to simulate the welding process and predict the quality of the weld before the actual fabrication process takes place. Indeed, a digital twin is defined as a real-time connection between a physical system and its virtual twin in order to simulate the system in real-time [88]. Such digital twin technology allows the engineer to simulate the weld pool, thermal fields, and distortions in the structure during the actual fabrication process in a more efficient way. Apart from this, multi-modal AI systems, which incorporate information from visual, acoustic, and electrical signals, are being designed to carry out automated weld defect detection processes. These systems have proved to be highly accurate in identifying various types of weld defects by employing machine learning algorithms on a large volume of welding signals and images [1]. These technologies enable continuous process monitoring and automated quality assurance.

Another notable technology in this field is human-centric collaborative welding systems. Industry 5.0 is based on human-machine collaboration, where human

expertise is combined with intelligent machines, enabling humans to monitor and interact with robotic welding machines through sophisticated interfaces. Collaborative robots, or cobots, help human welders carry out complex operations by performing repetitive tasks, thus enhancing productivity and safety. On a broader scale, the evolution depicted in Figure 3 clearly reflects a trend of moving from completely manual welding processes to highly intelligent cyber-physical manufacturing technologies. Each technology level represents an evolution from the previous one, from manual craftsmanship to automation, digitalization, artificial intelligence, and ultimately smart manufacturing technologies.

VII. Machine Learning Based Welding Defect Detection

Automated weld inspection is an essential research area that has gained importance due to the need for high reliability in industrial constructions such as pipelines, pressure vessels, and aircraft parts. Welding defects such as porosity, crack, slag inclusion, lack of penetration, and undercut may result in reduced structural integrity and may lead to catastrophic failure. The traditional weld inspection methods such as radiographic testing, ultrasonic testing, and visual inspection are effective but may need high human intervention and expertise. Machine learning (ML) methods have gained prominence for the automation of weld defect detection and classification tasks. The conventional machine learning-based approaches use hand-crafted feature extraction techniques. Then, classification algorithms are used. In the conventional approaches, features are extracted from the images of the weld defects by using image processing techniques such as gray-level co-occurrence matrices, histogram-based features, wavelet transforms, and texture features. Then, these features are used in classification algorithms such as Support Vector Machine, Random Forest, k-Nearest Neighbors, and Artificial Neural Networks. The effectiveness of these conventional approaches in classifying the weld defects from images obtained from radiographic/visual inspection has been demonstrated in several literatures.

Support vector machines have been applied to radiographic inspection after texture features are extracted from X-ray images, while random forests and related ensembles combine multiple decision rules. Process-monitoring studies also use arc voltage, acoustic emission, and thermal signals to identify deviations from nominal operation. These classical methods remain useful, especially with small or structured datasets, but their performance depends strongly on hand-engineered features. A feature set developed for one material, process, or sensor configuration may not transfer to another. They can also struggle when relevant patterns are spatially complex or when dimensionality grows faster than the available labeled data.

This led to the development and utilization of deep learning techniques that can learn hierarchies of features

directly from the data. Deep learning techniques have the ability to overcome the requirement of having to engineer features and can learn complex spatial and contextual information from the weld images and sensor data as shown in Figure 4 [1], [64].

VIII. Deep Learning Based Welding Defect Detection

Deep learning methods have revolutionized the field of automated weld defect detection by allowing models to learn representations of data directly from raw data. Traditionally, in conventional machine learning methods, features are defined manually, but deep learning models are able to learn features directly from data by applying multiple non-linear transformations to data. Deep learning models have achieved great improvement in weld defect classification accuracy. Convolutional Neural Network (CNN) is a type of deep learning model that is most frequently utilized in weld defect detection tasks. CNN models are comprised of convolution, pooling, and fully connected layers, where convolution and pooling layers are utilized to detect spatial features, and fully connected layers are utilized for classification purposes. CNN models have demonstrated superior performance in weld defect detection from various data sources such as radiographic images, optical inspection images, and thermal images. Several CNN studies report high within-dataset classification performance, but the values are strongly dependent on dataset construction, train-test separation, defect prevalence, and image source. This review therefore does not present an uncited “higher than 95%” value as a general welding result; study-level metrics must be interpreted with their dataset and validation protocol [1], [65].

In order to further enhance their performance, transfer learning methods with pre-trained models such as ResNet, VGGNet, and EfficientNet have been explored and utilized in weld inspection systems. This reduces the time required to train deep learning models with smaller labeled image datasets. Object detection models such as Faster R-CNN and YOLO have also been explored and implemented in weld inspection systems. These models are capable of detecting and localizing defects in weld images by identifying bounding boxes around defect regions in the images. This is useful in industrial inspection systems where localization of defects is required for automatic repair of such defects.

However, in spite of their high performance, there are a number of challenges associated with the implementation of deep learning models as well. One of the major challenges facing the implementation of these models is that they require a significant amount of labeled data for training. However, in industrial welding environments, the availability of labeled data might be a major issue as annotating the data might be a costly process. Also, deep learning models are known to be black boxes; hence, there might be a major problem in understanding the decision-

making process of these models. Such a situation might be a major issue in aerospace and nuclear manufacturing environments. Conventional CNN-based models may be more focused on visual features and might not be able to handle contextual knowledge. There has been a significant need for integrating these deep learning models with a knowledge system that has the ability to reason about the process and the defects in the welding process.

These challenges have led to the exploration of the potential of large-scale foundation models and large language models (LLMs) to integrate visual information, sensor signals, and domain knowledge to enable intelligent decision-making in welding inspection systems [1], [65].

A. Reinforcement Learning in Welding

Reinforcement learning (RL) has been investigated for adaptive welding control and related optimization tasks using multi-sensor states and reward-driven policies. The evidence spans recorded laser-welding data combined with simulated control, experimental friction-stir-welding optimization, and rail-welded-joint grinding; these validation settings are not interchangeable and are treated as conventional AI evidence rather than LLM evidence [73], [75]–[77].

1) Laser Welding: Complete Intelligent Control Architecture: The architecture of Günther et al. combines representation learning, temporal-difference prediction, and actor-critic control [73]. The reported monitoring setup uses camera images up to 1500 Hz at 144×176 pixels and three photodiodes sampled at 40 kHz over 1100–1800 nm, 400–600 nm, and 1050–1080 nm (source Section 2.1). A stacked denoising autoencoder compresses the images to 16 features. On eight test datasets it reports 16.6% mean reconstruction error versus 15.5% for PCA; Table 2 reports $0.81 \pm 0.11\%$ SVM classification error for the 16-feature autoencoder representation versus $1.79 \pm 0.19\%$ for PCA [73]. These are representation and classifier results, not end-to-end autonomous welding accuracy.

The 16-dimensional representation plus three photodiode channels forms a 19-dimensional state for “nexting” prediction. Figure 6 of the source reports a distance error reduction from 0.48 after one training pass to 0.28 after convergence on an unseen process. The paper reports 82% four-class prediction accuracy and 93.1% accuracy only when the task is collapsed to acceptable versus insufficient weld regions; it does not report the 94% value previously stated here [73]. The actor-critic controller was then evaluated in a preliminary laser-welding simulator over 30 independent runs. The simulated power range was 1.5–5 kW, and the reported computation time was 0.34 ms per control iteration, but the authors explicitly call for additional industrial validation [73].

2) Friction Stir Welding: Multi-Objective Real-Time Optimization: Model-free deep RL (DQN and PPO) uses in-situ sensor fusion of spindle torque, axial force, interface temperature, and tool vibration [75]. The state

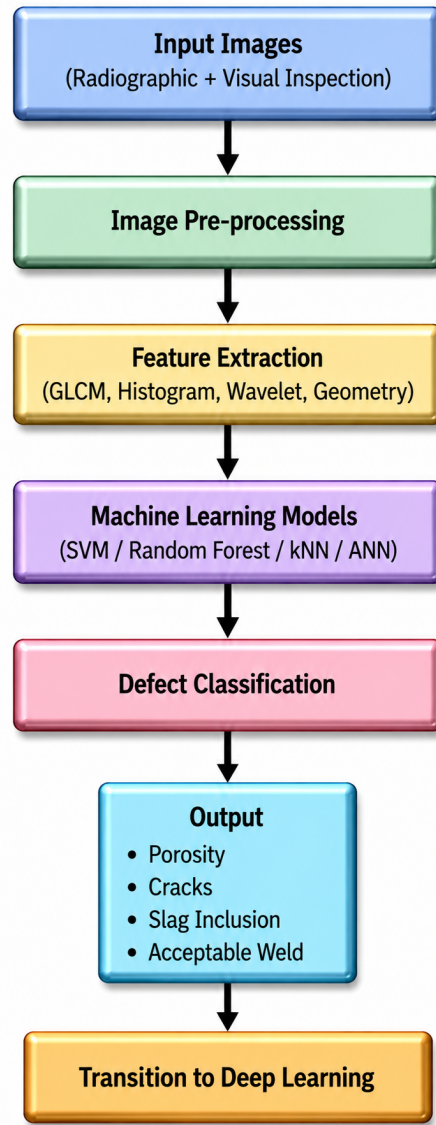


Fig. 5. ML based Welding Defect Detection

space encodes thermo-mechanical conditions while continuous actions optimize tool rotational speed (700-1200 rpm), traverse speed (30-70 mm/min), and shoulder-to-pin diameter ratio (SP). A sophisticated multi-objective reward function simultaneously minimizes intermetallic compound (IMC) thickness, thermal variance, and flash formation while maximizing tensile strength and electrical conductivity.

Experimental validation on AA6061-T6/Cu butt joints reports a 19.7% tensile-strength increase (142 ± 5.8 MPa to 170 ± 4.2 MPa), a 23.3% reduction in intermetallic-compound thickness (8.6 ± 0.7 μm to 6.6 ± 0.5 μm), and a 17.2% improvement in electrical conductivity relative to the optimized static baseline [75]. These values are stated in the article abstract. Additional disturbance amplitudes, recovery times, and action percentages were excluded

because they could not be traced reliably to accessible primary-source text.

3) Rail Grinding and Process Extensions: Multi-objective RL also extends to rail-welded-joint grinding using a dual-critic Deep Deterministic Policy Gradient framework [76]. The source reports 55 field-measured rail-weld profiles, sampled over 5 m segments at 1 mm intervals (5001 points per profile). After 6000 training steps, the actor and critic losses converged. Across 300 sampled trade-off weights the method generated more than 220 feasible strategies per case, and the reported case result improved profile-trough restoration from -0.41 to -0.18 mm [76]. This is rail-maintenance optimization, not 6000 physical welding experiments.

Additional extensions include closed-loop Q-learning for laser power modulation (20-26 line convergence) [74]

and hierarchical RL for multi-robot welding flow-shops, demonstrating RL’s scalability across welding domains.

4) RL Contributions to Defect Detection and Prevention: Beyond parameter control, RL contributes significantly to defect mitigation through three mechanisms:

Indirect Defect Prevention via Process Stabilization: The laser-welding study demonstrates that a 19-dimensional state (16 image features plus three photodiode channels) can support immediate and temporally extended quality prediction. Its five-step-ahead example is a forecast of expert-assigned quality labels, not direct evidence that pores were prevented in physical closed-loop trials [73]. The FSW study reports sensor-informed adjustment of process parameters and improved joint properties, but remains a conventional RL system [75].

Prediction-error monitoring: Temporal-difference errors can, in principle, indicate mismatch between expected and observed signals. However, the cited laser-welding study does not report 87% sensitivity or 92% specificity for anomaly detection; those values have been removed and would require a separately defined anomaly benchmark [73].

Quality proxy prediction: Expert labels based on EN ISO 13919-1:1996 were encoded from 1 (insufficient) to 4 (ideal). For an unseen process, Figure 6 of the source reports distance error 0.48 after one pass and 0.28 after convergence; the associated classification error is 0.18, corresponding to 82% four-class accuracy, while a binary acceptable/insufficient grouping reaches 93.1% [73].

5) Fundamental Limitations Necessitating LLM Integration: Despite these advances, RL’s core limitations render it insufficient for comprehensive weld defect detection and interpretation:

Sample complexity: The rail-grinding study reports convergence after 6000 training steps, not 6000 welding episodes. It does not report a 30-minute physical-weld duration or a 3000-hour training campaign [76]. Physical deployment of exploratory policies would nevertheless require simulation, offline data, or other safety-preserving training strategies.

High-dimensional sensing: The cited architecture compresses camera data and combines the representation with 40-kHz photodiode channels because raw multimodal streams are impractical for direct online learning. The source does not demonstrate a specific missed micro-crack threshold or attribute aliasing failures to tile coding, so those quantitative assertions are omitted [73].

Non-stationary rewards: Multi-objective rewards can change with material batch, thermal history, tool condition, and the relative weighting of quality and productivity. Activation-energy variation and Pareto-instability ranges could not be traced to the cited FSW source and are therefore not reported as experimental results.

Exploration risk: Exploratory actions can produce poor process outcomes, which is why learning directly on safety-critical equipment requires bounded actions and validated

simulators. The cited laser-welding paper does not report a 12% defect rate or the contamination/power example; neither is treated as experimental evidence [73].

Absence of Causal Defect Reasoning: RL predicts quality proxies but cannot classify “transverse crack vs. longitudinal porosity” or generate interpretable mitigation strategies. Zero-shot generalization to novel defects (zinc coating vaporization, oxide inclusions) fails completely [73].

These limitations motivate combining real-time perception and control with language-grounded retrieval and explanation, but they do not establish that an LLM itself can consume raw streams or control the process. No matched welding experiment supports a 10^3 -fold sample-reduction claim. In the architecture developed later in this review, conventional edge models and certified controllers retain sensing and actuation, while the LLM supports retrieval, explanation, planning, and human interaction.

IX. LLM Fundamentals

A. Fundamentals of LLMs

The evolution from Industry 4.0 to Industry 5.0 has heightened the need to process vast amounts of heterogeneous and mostly unstructured industrial data. Within this evolution, Large Language Models (LLMs) have been studied as language-capable foundation models within the broader family of large-scale pretrained artificial intelligence systems. Foundation Models (FMs) are a category of large-scale neural networks that are trained using vast amounts of text, code, and multimodal data utilizing self-supervised learning approaches. This training allows Foundation Models to be fine-tuned or learn utilizing prompts to perform a wide variety of tasks [29].

Unlike traditional machine learning architectures, which are normally fine-tuned for specific objectives such as defect classification or parameter estimation, LLMs have emergent properties that are largely a consequence of the scale and data variety. These properties include reasoning, abstraction, and generalization. Examples of these architectures are Generative Pre-trained Transformer (GPT) and variants thereof, which are best suited for generative and reasoning tasks, and BERT (Bidirectional Encoder Representations from Transformers), an architecture optimized for contextual understanding and classification. In the context of manufacturing systems, these properties are represented by two primary capabilities. Firstly, LLMs are capable of text generation and semantic understanding by modeling complex linguistic structures, context, and even latent reasoning. This enables the ingestion and subsequent structuring of unstructured data from the industrial domain, such as maintenance records, standard operating procedures, inspection results, and even operator feedback. For instance, an LLM can interpret operator feedback in the form of observations about process behavior, such as reports of abnormal vibration,

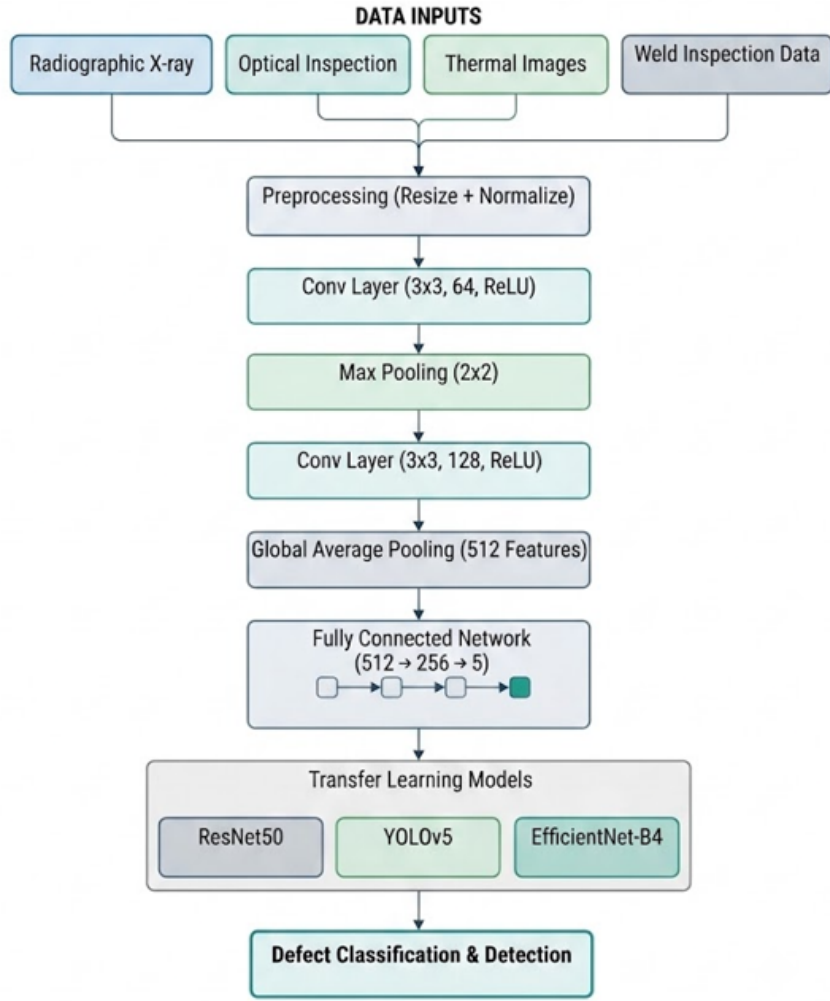


Fig. 6. Deep learning pipeline for weld defect detection

and connect these to known failure histories or process upsets [16].

Secondly, contemporary LLMs have been found to have robust code generation and automation capacities owing to their training on large repositories of programming languages. This enables them to generate, debug, and even explain control logic in industrial control systems, data processing code, as well as automation code. This helps in reducing the entry barriers to advanced manufacturing automation since users, even without expertise, can interact with complex manufacturing systems via natural language. This, therefore, promotes a more inclusive and human-centric manufacturing environment [29].

Beyond natural language processing techniques, recent breakthroughs have seen the development of Vision-Language Models (VLMs), which combine visual and textual information in a unified representation space. VLMs, such as CLIP (Contrastive Language-Image Pre-training) and LLaVA (Large Language and Vision Assistant), integrate image features from cameras or sensors with

language. This helps in achieving zero-shot and few-shot reasoning. This, therefore, enables manufacturing defect identification or monitoring purely based on semantic descriptions, even in the absence of task-specific labeled data. [30].

B. LLM Architectures

The performance and scalability of contemporary LLMs are underpinned almost exclusively by the Transformer architecture, originally introduced by Vaswani et al. in 2017. This architecture marked a decisive departure from sequential modeling approaches such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, enabling efficient parallel computation through self-attention mechanisms [30].

At the core of the Transformer lies the self-attention mechanism, commonly expressed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

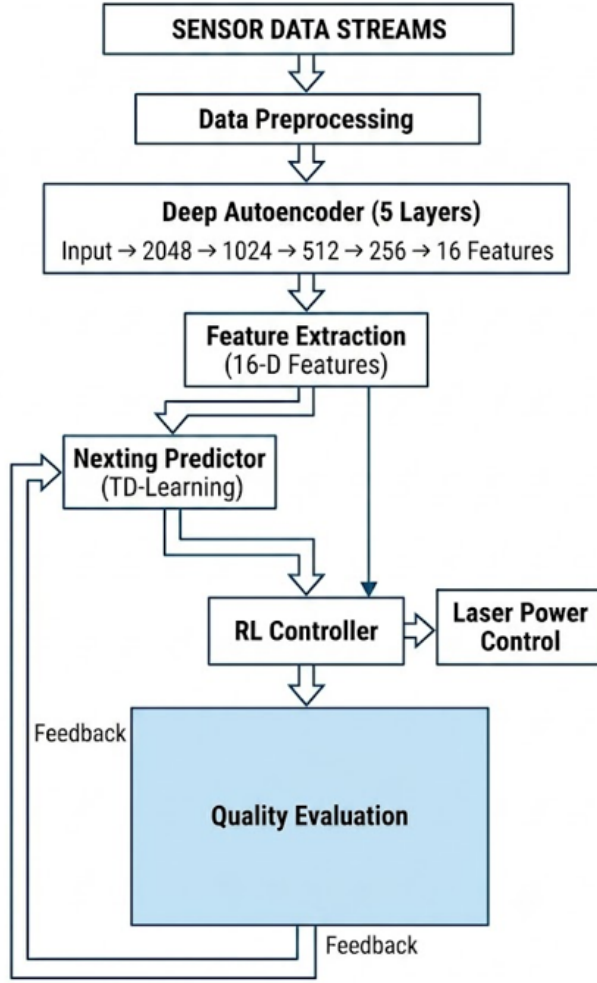


Fig. 7. Complete RL architecture for Welding

where queries (Q), keys (K), and values (V) are learned projections of the input tokens. This formulation allows the model to dynamically weight the relevance of each token relative to all others, independent of their positional distance. In manufacturing contexts, such global dependency modeling is particularly valuable, as it enables the association of information dispersed across lengthy technical documents, process logs, or safety manuals—relationships that sequential models often fail to capture [30].

Several Transformer-based architectural variants are commonly employed in industrial and manufacturing-oriented applications. Encoder-only models, such as BERT, process input sequences bidirectionally to generate contextual representations and are used for tasks including defect-report classification, document retrieval, and supplier-communication analysis [85]. Decoder-only models, such as GPT, processes input data in an autoregressive manner and is particularly useful for generative tasks such as report drafting, conversational assistants,

and synthetic data generation [29]. Vision Transformers (ViTs) extend the same attention-based mechanism for processing visual data as well by representing images as a sequence of fixed-size patches; hence, this type of global reasoning for processing visual data forms the basis of all modern industrial vision-language systems [30].

Overall, the architectural flexibility and representational capacity of Transformer-based LLMs position them as a foundational enabler for human-centric, intelligent, and resilient manufacturing systems envisioned under the Industry 5.0 paradigm as shown in Figure 7.

The architecture of the proposed industrial Large Language Model (LLM) system depicts the flow of data within the model and the manner in which heterogeneous data from the manufacturing domain is transformed into intelligent information. The input stage of the model involves the transformation of the raw data into numerical tokens that can be processed by the neural network architecture. The numerical tokens are fed into the Transformer backbone of the architecture, which acts as the core

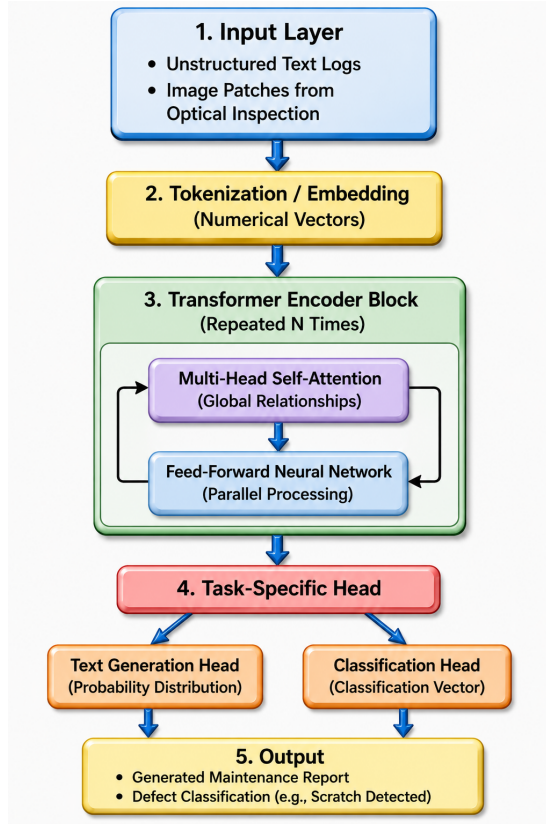


Fig. 8. Example of a Generic LLM System Architecture

computational engine for the model. The core engine consists of a stack of multi-head self-attention and feed-forward neural network layers that enable the model to process complex contextual relations among all the tokens in the input data in parallel. The processed tokens are directed to the output heads for the performance of tasks such as text generation and classification for defect detection and quality evaluation tasks [30].

C. Multi-Agent and LLM Synergy

In complex manufacturing systems, the complexity and interconnectedness of tasks may be beyond the scope that a single intelligent system can manage. This has prompted the employment of a Multi-Agent System (MAS), in which multiple autonomous agents collaborate to accomplish a common goal, with LLM playing a central role as a cognitive agent. In this system, the role of the LLM is to allow the agent to understand high-level human instructions and execute them in a coordinated fashion in a distributed manufacturing system [16].

A key capability underpinning this paradigm is instruction following and task decomposition. Unlike traditional industrial robots that rely on rigid, pre-programmed logic, LLM-enabled agents can process abstract and goal-oriented commands. For example, when an operator issues a request to analyze production bottlenecks and propose scheduling adjustments, the LLM agent can autonomously

decompose this instruction into a sequence of subtasks, such as querying a digital twin for cycle-time data, identifying performance-limiting stations, generating an alternative schedule, and summarizing the results for human review. This semantic reasoning capability effectively bridges the gap between human intent and machine execution, enabling more flexible and human-centric automation. One of the most important features that make this system possible is the ability to perform instruction following and task decomposition. Unlike other industrial robots that use a fixed logic system in manufacturing systems, the instruction following and task decomposition feature allows the LLM agent to perform tasks using high-level and goal-oriented human instructions. For example, in a manufacturing system, when a human agent gives a command to perform a task that involves analyzing the bottlenecks in the production system and generating a new schedule for the system, the LLM agent can independently decompose this high-level human instruction into a series of tasks that need to be executed in a particular sequence to accomplish the task on [16].

1) Knowledge-Enhanced Reasoning: A major challenge associated with deploying LLMs in safety-critical manufacturing contexts is hallucination, whereby models generate responses that appear plausible but are factually incorrect. Retrieval-augmented generation (RAG) can reduce

this risk by supplying external domain knowledge at inference time, although retrieval does not guarantee factual correctness. For example, CausalKGPT combines an LLM with a causal quality-related knowledge graph so that defect hypotheses can be linked to documented cause–effect relationships rather than produced solely from parametric memory [16].

2) Cross-Modal Collaboration: Modern industrial MAS can assign different data modalities to specialized agents. A visual agent equipped with a vision–language model such as LLaVA may describe an observed component, while a knowledge-oriented agent retrieves acceptance criteria from technical documentation. Their outputs can then be compared with formal specifications before being shown to a human inspector. This is a transferred quality-assurance pattern rather than demonstrated welding performance [30].

3) LLM Applications in Industry 5.0: Industry 5.0 extends the efficiency-oriented goals of Industry 4.0 to include human-centered design, sustainability, and resilience. Within this setting, an LLM can serve as a language interface between human objectives and planning software, provided that its outputs remain constrained by domain knowledge and human approval. Logistics studies illustrate the transferred mechanism: language models can summarize heterogeneous information such as weather, supplier communication, and disruption reports to support anticipatory planning [29]. This evidence motivates analogous decision-support research in welding but does not demonstrate welding performance.

4) The Human–AI Interface and Digital Twins: One of the most significant contributions that LLMs will make in the context of Industry 5.0 is the improvement of the interface and interaction with cyber-physical systems and digital twins. Natural language interfaces enable operators to ask digital twin models questions, removing the need to explore complicated digital interfaces and low-level system log information. For instance, operators can ask LLMs to explain the reason for unexpected robot stoppages, and the LLM will be able to access the digital twin to obtain the relevant sensor data and system logs to give the operator a simple and easily understood explanation. This reduces the cognitive burden for operators to interact with advanced automation systems [33].

Another significant contribution is the codification of expert knowledge, specifically in areas like additive manufacturing, where quality is often judged qualitatively. The use of LLMs and vision-language models will enable the codification of expert knowledge and the scaling of expert-level screening and qualification processes over entire production lines, effectively increasing human expertise [85].

5) Defect Detection and Quality Assurance: LLMs are also transforming the field of industrial quality assurance by enabling the use of semantic defect detection. For instance, whereas traditional automated optical inspection

systems may only be able to identify known classes of defects, the use of LLMs enables the identification of rare classes of defects with the help of the semantic description provided to the system. In this setting, the use of such systems with the help of causal reasoning frameworks enables the system not only to identify defects but also to provide suggestions for process optimization to avoid the occurrence of such defects in the future. This is a major step towards the development of intelligent and self-improving quality systems in Industry 5.0 manufacturing plants [16], [64].

D. LLM and RAG for Welding Defect Detection

A fundamental challenge in deploying LLMs for welding defect detection in industrial environments is the risk of hallucination—the generation of responses that are linguistically fluent but factually incorrect or ungrounded in validated domain knowledge. In safety-critical applications such as aerospace or nuclear component fabrication, an incorrect defect diagnosis can lead to structural failures with catastrophic consequences. Retrieval-Augmented Generation (RAG) addresses this limitation by decoupling the parametric knowledge of the LLM from the factual content it generates: at inference time, the model retrieves relevant verified documents—welding procedure specifications, standards such as ISO 5817 and AWS D1.1, historical inspection records, and FMEA knowledge bases—and conditions its output on this retrieved context [17], [82].

1) Architecture and Mechanism of RAG in Welding Contexts: A RAG pipeline for welding defect detection typically comprises three tightly coupled components. First, a document ingestion and indexing module processes heterogeneous welding documents—including procedure qualification records, nondestructive testing reports, metallurgical assessments, and operator logs—into semantically rich vector embeddings stored in a dedicated knowledge store. Second, when a query is submitted (for example, a sensor anomaly report or a defect image description), a dense retrieval module identifies the most semantically relevant passages from the knowledge store using approximate nearest-neighbor search over embedding space. Third, the LLM receives both the original query and the retrieved context as a combined prompt, generating a response that is grounded in verified procedural and metallurgical knowledge rather than relying solely on parametric memory [35], [83].

This architecture is particularly well-suited to welding defect detection because the defect–cause relationships in welding are highly process-specific, material-dependent, and governed by standards that evolve over time. Parameterized LLM knowledge, fixed at training time, cannot reliably represent such dynamic and specialized content. RAG enables the system to remain current as standards are revised, as new alloys are introduced, and as produc-

tion processes are modified—without requiring expensive model retraining [17].

2) RAG-Augmented Root Cause Analysis: One of the most impactful applications of RAG in welding quality assurance is root cause analysis (RCA). Conventional RCA relies on expert engineers manually correlating defect observations from non-destructive testing reports with process parameter logs, material certificates, and historical failure records—a time-consuming and error-prone process that depends heavily on individual expertise. RAG-augmented LLMs automate this correlation by retrieving and synthesizing relevant evidence from multiple knowledge sources simultaneously [16].

CausalKGPT integrates an LLM with a causal quality-related knowledge graph constructed from aerospace-manufacturing defect surveys [16]. It demonstrates how retrieval can expose documented cause-effect paths during root-cause analysis. Transferring that mechanism to welding could help compare competing explanations for porosity or cracking, but the cited CausalKGPT evaluation does not validate laser-welding diagnosis; welding-specific evidence is provided separately by Wang et al. [35].

In a complementary framework for bridge weathering-steel welding, an LLM combines multi-perspective associative memory, hybrid retrieval, task decomposition, and expert-guided verification to analyze defect causes [35]. The source reports improvements in ROUGE-L, BERTScore, and expert evaluation across the tested models. Claims of more than 12,000 inspection records or superior consistency to experienced engineers are not used here because no accessible primary-source passage specifying the dataset and comparison protocol was located.

3) FMEA Knowledge Graphs and Graph-RAG: Failure Mode and Effects Analysis (FMEA) is a structured risk assessment methodology widely used in welding-intensive industries, including automotive and aerospace manufacturing. Traditional FMEA documents are maintained as static spreadsheets or databases that require continuous manual updates as processes evolve. Graph-RAG frameworks address this limitation by representing FMEA knowledge as dynamic, interconnected knowledge graphs in which nodes correspond to failure modes, effects, and causes, while edges encode causal and conditional relationships with associated severity and detectability scores [17].

When an LLM receives a defect query in a Graph-RAG system, the retrieval module traverses the knowledge graph to identify the subgraph most relevant to the observed failure mode, providing the LLM with a structured and semantically rich context for diagnosis. This graph-structured retrieval outperforms flat document retrieval in scenarios involving multi-factorial defect interactions—for example, where porosity results from the concurrent influence of improper interlayer temperature, insufficient gas pre-purge, and excessive travel speed, none of which individually would trigger a threshold-based alert [17]. By

retrieving and reasoning over the entire causal subgraph, the LLM can provide a holistic explanation of the defect mechanism and recommend multi-parameter corrective actions.

4) Vision-Language RAG for Multimodal Inspection: The extension of RAG frameworks to multimodal settings—combining visual inspection data with textual knowledge—represents a frontier capability for welding defect detection. In Vision-Language RAG systems, image embeddings from weld inspection photographs or radiographic images are projected into the same semantic space as text embeddings from the knowledge store, enabling cross-modal retrieval: a visual query (an anomalous weld bead image) retrieves both visually similar historical cases and textually relevant procedural knowledge describing the associated defect type and corrective actions [81], [85].

In additive manufacturing quality assurance, vision-language models augmented with RAG have been evaluated for zero-shot qualification of build layers by retrieving acceptance criteria from technical standards in response to visual anomalies [85]. This mechanism is relevant to welding because base materials, joint geometries, and acceptance criteria vary across applications. A maintained knowledge base may allow a vision-language RAG system to retrieve updated standards or related defect cases without retraining the underlying model; this remains a research hypothesis until validated on welding-specific distribution shifts [81], [83].

5) Limitations and Open Challenges: Despite their significant promise, RAG-augmented LLM systems for welding defect detection face several unresolved challenges. First, the quality of the retrieved context is critically dependent on the completeness and consistency of the underlying knowledge base; sparse or inconsistently labeled inspection records will degrade retrieval relevance and, consequently, diagnostic accuracy [82]. Second, the latency introduced by retrieval operations may conflict with real-time inspection requirements, particularly in high-throughput robotic welding cells where decision cycles are measured in milliseconds [17]. Third, knowledge graph construction and maintenance require significant domain expertise and ongoing curation effort, representing a barrier to adoption for smaller manufacturers without dedicated quality engineering resources [16]. Addressing these challenges through advances in retrieval efficiency, automated knowledge graph updating, and uncertainty-aware generation represents a rich agenda for future research.

X. LLM-Enabled Smart Welding

A. Conceptual Framework

LLM-enabled smart welding refers to the integration of Large Language Models (LLMs), multimodal LLMs, or VLMs into welding knowledge, inspection, planning, and human-interaction workflows. Current evidence supports decision assistance, offline analysis, and prototypes; it does

not establish end-to-end autonomous industrial welding. Embodied-robot literature is therefore treated as transferred architectural motivation rather than direct welding validation [19].

Within this framework, the welding system is conceptualized as a layered embodied system rather than as an LLM acting alone. The physical body and deterministic controller interact with materials; perception models convert sensor data into estimates; and the LLM supports multi-domain knowledge integration, semantic interpretation, and high-level reasoning. This separation prevents open-set visual recognition or generic embodied-robot results from being presented as demonstrated welding performance [19], [81].

The conceptual architecture of LLM-enabled smart welding can be structured as a four-layer hierarchical framework designed to optimize data flow, knowledge utilization, and decision-making across the welding life-cycle [33].

- **Physical Layer:** This layer comprises the tangible components of the welding system, including robotic manipulators, welding power sources, and an array of intelligent sensors such as RGB-D cameras, acoustic emission sensors, and thermal imaging devices. These elements capture the real-time physical state of the welding process, including arc behavior, weld pool dynamics, and thermal distribution, thereby forming the primary source of multimodal data for higher-level reasoning [33].
- **Connection Layer:** The connection layer carries data between the shop floor and computational infrastructure. Industrial Internet of Things (IIoT) protocols such as MQTT and OPC UA can transport typed messages, while wired industrial networks or suitably engineered wireless links provide the latency and reliability required by the application. Safety-critical control should not depend on an unconstrained LLM response or a best-effort network path [33], [81].
- **Virtual Layer:** At the virtual layer, Digital Twin (DT) models represent the physical welding cell at a fidelity appropriate to the decision. The DT, rather than the LLM, executes predictive simulations and “what-if” analyses. An LLM may formulate a query or summarize results, but parameter changes require validation by the physics model, operating envelope, and deterministic safety logic before physical execution [33].
- **Supporting Layer:** The supporting layer contains the LLM or MLLM, retrieval components, provenance tracking, and interfaces to validated analytical models. It receives textual records and encoded summaries rather than uncontrolled raw sensor streams. Its appropriate role is high-level retrieval, explanation, and planning support; adaptive control remains with validated controllers, and recommendations shown in Table IV require human review [33], [81].

TABLE VI
Comparison of traditional and LLM-enabled welding systems

Aspect	Traditional welding systems	LLM-enabled welding systems
Decision-making paradigm	Rule-based and operator-dependent, relying on predefined procedures and expert intuition	Knowledge-driven and adaptive, supported by LLM-based reasoning and contextual understanding
Process parameter selection	Empirical tuning and trial-and-error experimentation	Predictive parameter optimization using historical data, physics-based models, and semantic reasoning
Human-machine interaction	Limited to manual inputs and static interfaces	Natural language interaction enabling intuitive, human-centric control and feedback
Defect detection and analysis	Reactive inspection with predefined defect classes	Semantic defect identification and causal analysis enabling proactive defect mitigation
Adaptability to variability	Low adaptability to material, geometry, or environmental variations	High adaptability through continuous learning and contextual inference
Knowledge utilization	Tacit knowledge confined to experienced operators	Explicit codification and scalable reuse of expert knowledge across systems
System transparency	Limited interpretability of automated decisions	Explainable decision support enhancing trust and operator understanding
Alignment with Industry 5.0	Primarily efficiency- and automation-centric	Human-centric, resilient, and sustainability-oriented

B. Direct Evidence Corpus

Table VII identifies the 21 publications that directly address a welding-specific task using a language-capable LLM, multimodal LLM, VLM, RAG system, LLM-mediated agent, or an explicitly identified LLM represen-

tation component. Fifteen are evaluated or comparative studies (E1), five are prototypes or preliminary validations (E2), and one is a focused secondary review (E3). Conventional welding AI and transferred manufacturing studies remain in this review because they supply essential sensing, control, materials, safety, and architectural context, but they do not contribute to the direct count.

The evidence supports five present clusters: welding-knowledge retrieval and Q&A [35], [90], [92], [95]; multi-modal defect assessment and reporting [13], [91], [93], [94], [96], [99], [102]; data-sparse equation discovery [78]; LLM-initialized numerical prediction [14]; and high-level robot or process planning with conventional execution layers [48], [98], [101], [103]. Under a strict definition requiring routine plant operation, a documented safety case, real-time closed-loop actuation, and evaluation across production variability, the industrial-autonomy count is zero.

C. LLM for Welding Parameters

Large Language Models (LLMs) are increasingly employed to recommend optimal welding parameters—such as voltage, current, and travel speed—by synthesizing information from technical standards, historical experimental datasets, and scientific literature. By leveraging their ability to integrate heterogeneous knowledge sources, LLMs enable parameter selection strategies that are both data-driven and context-aware, moving beyond purely empirical tuning approaches [15], [78].

Conventional machine learning methods often encounter limitations in welding applications because complex processing routes, involving sequential stages of heating, deformation, and cooling, generate high-dimensional and sparse representations when encoded as tabular data. Such representations are difficult to generalize across processes and materials. LLMs address this challenge by treating welding and processing routes as structured text sequences, thereby capturing the semantic meaning of operations such as “quenching,” “annealing,” or “tempering” in a manner that aligns more naturally with expert reasoning [15].

A prominent example of this approach is the SteelBERT materials language encoder, which employs a disentangled attention mechanism to map complex fabrication sequences to resulting mechanical properties with high predictive accuracy. By preserving both procedural context and numerical information, SteelBERT demonstrates how language-based representations can bridge the gap between process descriptions and material performance [15]. Complementary advances have been reported by researchers at The Pennsylvania State University, who developed the T2EGPT (Text-to-Equations Generative Pre-trained Transformer) framework to predict humping onset in high-speed laser welding. This framework illustrates the potential of LLMs to connect linguistic process descriptions with underlying physical phenomena [78].

Key capabilities enabled by LLM-based parameter optimization frameworks as shown in Table V include:

- **Physics-informed equation discovery:** Frameworks such as T2EGPT utilize LLMs to derive dimensionless physical relationships from sparse experimental data and domain literature. These relationships, consistent with governing non-dimensional numbers such as the Weber number, enable the model to balance inertia-driven melt flow against capillary-driven surface stabilization in high-speed welding processes [78].
- **Numerical semantic preservation:** Specialized encoders such as SteelBERT are trained to represent materials terminology and numerical context, including temperatures and deformation magnitudes. The cited study evaluates material-property prediction rather than welding-process outcomes, so it is used here as transferred evidence for domain-specific representation [15].
- **Fine-tuning with limited datasets:** As transferred materials evidence, SteelBERT was fine-tuned on 64 experimental samples of 15Cr austenitic stainless steels. The source reports cross-validated R^2 values of $89.85 \pm 6.17\%$ for yield strength, $88.34 \pm 5.95\%$ for tensile strength, and $87.24 \pm 5.15\%$ for elongation, and then used the model to propose additional cold rolling and tempering steps (source Fig. 8 and accompanying text) [15]. These results motivate small-data welding hypotheses but are not welding-parameter experiments and are not counted as direct evidence. A newly indexed direct study combines an LLM-based embedding initializer with a wavelet irregular Transformer and differentiable Gumbel sampling to predict deformation and residual-stress fields in thin-plate welding [14]. The work reports validation on dedicated simulation data supplemented by physical pulsed-laser-welding experiments. It is included as E1 because an LLM component is explicitly part of the welding predictor, but the evidence boundary is narrow: WITGS is primarily a numerical spatiotemporal model, and the paper does not demonstrate natural-language decision making or closed-loop welding control.

D. LLM for Programming Welding Robots

Large Language Models can function as code-generation and high-level orchestration components, potentially lowering the barrier to configuring robotic welding systems. Direct welding prototypes exist, but natural-language convenience does not establish safe execution or eliminate the need for robot programming expertise, simulation, and approval [48], [101], [103].

Within a Robot Digital Twin (RDT) environment, LLM-assisted programming can connect an operator’s high-level request to a structured scene representation [33]. Vision and physical sensors, such as 3D cameras and force sensors, first estimate the workpiece geometry and

TABLE VII
Direct LLM/VLM/RAG evidence in welding ($n = 21$)

Ref.	Welding task	Model/system	Validation boundary	Class
[89]	Titanium-weld microstructure and defect workflow	General LLMs plus ConvNeXt	Image-processing/code-generation case; perception remains conventional vision	E1
[35]	Bridge-steel defect-cause analysis	LLM with memory, retrieval, and expert verification	Offline Q&A and expert evaluation; no process control	E1
[90]	Access to welding standards	Standards-grounded assistant	Design-case evaluation; normative interpretation remains human responsibility	E1
[78]	Laser-weld humping onset	T2EGPT equation discovery	Sparse experimental/literature data; scientific discovery, not control deployment	E1
[91]	Visual weld acceptability	GPT-4o/LLaVA with WeldPrompt	Real and online images; generalization and precision-recall trade-offs remain	E1
[92]	Ship-welding process Q&A	KG plus vector Hybrid-RAG	Welding-requirement benchmark; no physical execution	E1
[93]	Visual defect assessment	Gemma/Qwen VLMs versus YOLO	Image benchmark; inadequate for autonomous acceptance	E1
[94]	Defect localization and report generation	YOLO-Qwen vision-language fusion	Dataset evaluation; language is added to detector outputs	E1
[95]	Ship-welding decision support	Multi-agent hybrid retrieval	Question benchmark for traceable answers	E1
[96]	Image-based severity assessment	Zero-shot VLMs versus YOLO	Human-adjudicated subset; not reliable for standalone acceptance	E1
[97]	Resistance-spot-weld quality	Deep learning plus LLM	Process/performance evaluation; deployment evidence limited	E1
[98]	Welding-robot path planning	LLM-extracted knowledge plus planners	Grid simulation, not a production robot cell	E1
[99]	Defect detection and diagnostic reasoning	Multimodal WeldLLM	Dataset evaluation; external plant validation remains open	E1
[13]	Welding X-ray defect decision support	Visual-prompt-guided Weld-LLaVA	Peer-reviewed radiography assistant; qualified inspector remains responsible	E1
[14]	Thin-plate deformation and residual-stress prediction	LLM-initialized WITGS numerical predictor	Simulation plus physical experiments; language component is an initializer, not a controller	E1
[100]	Nuclear-welding standards management	LLM, RAG, vector database	Architecture/prototype; safety case not demonstrated	E2
[48]	Human-robot welding interaction	Welding LLM, VLM, speech, XR, ROS2	Simulated real-world scenarios; not certified operation	E2
[101]	Welding-island orchestration	TrajectoryGPT, WeldGPT, RobotGPT, OperationGPT	Module experiments/case studies; not continuous deployment	E2
[102]	Radiographic interpretation	VLM-assisted localization/explanation	Preliminary; inspector remains decision maker	E2
[103]	Shipyard procedure and robot-plan generation	Qwen-based WeldGPT, LoRA, RAG, digital twin	Historical data and preliminary offline results	E2
[104]	Welding defect detection	Focused VLM review	Secondary synthesis of an early field	E3

state. The LLM may then propose a behavior sequence or code for a request such as applying a fillet weld along a specified seam. A deterministic service wrapper must parse, simulate, constrain, and approve the proposal before any command reaches the physical robot; the wrapper also manages execution state, safety constraints, and synchronization between virtual and physical systems [19], [38].

Digital twins can generate synthetic scenes and labels for auxiliary perception tasks such as pose estimation and robotic gripping. An LLM may help specify scenarios or organize annotations, but the simulator supplies the

geometry and ground truth. The cited work is transferred robotics evidence; it does not quantify annotation savings or deployment speed in a welding cell [105].

E. LLM in Quality Monitoring

LLMs and multimodal LLMs are emerging as semantic layers for welding-quality workflows involving sensor summaries, inspection reports, images, and maintenance records. Direct studies show useful capabilities, but comparative evaluations also show calibration and generalization limitations; these models should support, not replace, qualified inspection [35], [91], [93], [96].

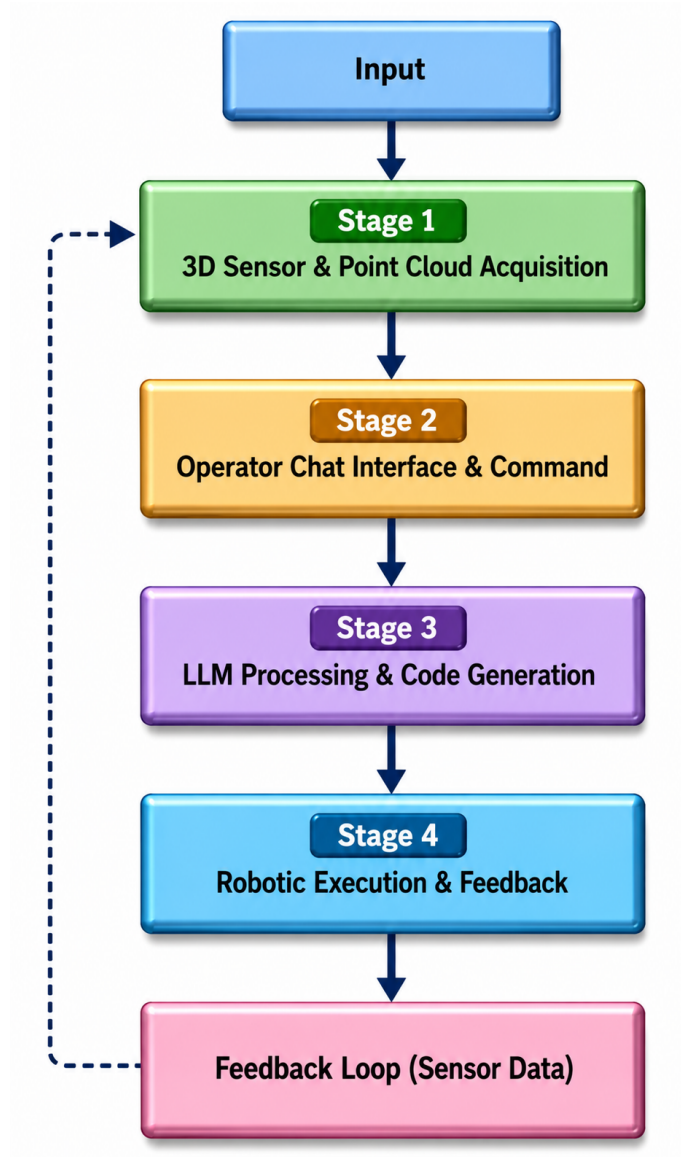


Fig. 9. Conceptual Flow Of LLM Robotic Welding

One prominent application is root cause analysis, exemplified by systems such as CausalKGPT. These systems integrate LLMs with structured causal representations, such as structure causal graph-based sum-product networks, to eliminate spurious correlations commonly found in manufacturing documentation. By reasoning over validated causal chains, CausalKGPT can identify how interacting process factors—such as mandrel runout and excessive forming pressure—contribute to complex failure modes, thereby supporting reliable and actionable diagnostics [16].

LLMs also enhance failure analysis and reporting through Retrieval-Augmented Generation (RAG) frameworks. By querying Failure Mode and Effects Analysis

(FMEA) knowledge graphs, LLMs can provide grounded responses to analytical queries, such as identifying the dominant causes of porosity at a specific process stage. This grounding significantly reduces hallucination and improves trustworthiness by anchoring generated explanations in verified knowledge graph entities [17].

Multimodal LLMs can also be evaluated through visual question answering for weld-quality assessment. Weld-Prompt compares prompting strategies on real and online weld images and shows that prompt choice changes the precision–recall balance; it does not establish independent certification of weld acceptability [91].

Weld-LLaVA extends this direct evidence base with a visual-prompt-guided vision–language assistant for weld-

TABLE VIII
LLM-driven parameter optimization metrics in smart welding systems

Metric	Description	Optimization relevance
Heat input stability	Measures the consistency of thermal energy delivered to the weld zone across time	Reduces microstructural variability, residual stresses, and distortion
Weld pool geometry	Characterizes bead width, penetration depth, and pool shape inferred from sensor data	Ensures consistent fusion quality and dimensional accuracy
Defect probability	Estimated likelihood of defects such as porosity, lack of fusion, or cracking	Enables proactive defect prevention through adaptive parameter tuning
Process energy efficiency	Ratio of effective joining energy to total energy consumption	Supports sustainable manufacturing and reduced operational cost
Arc or keyhole stability	Quantifies temporal fluctuations in arc length or keyhole dynamics	Improves process robustness and reduces defect formation
Parameter sensitivity	Assesses the influence of individual parameters on weld quality outcomes	Guides explainable and targeted parameter adjustments
Human workload index	Combines physical and cognitive load indicators of the operator	Aligns optimization with human-centric Industry 5.0 principles

ing X-ray defect decision support [13]. Because the article was registered shortly before the top-up search, it was not present in the initial OpenAlex/Semantic Scholar export. It is classified as E1, but its role is decision support: radiographic acceptance remains the responsibility of qualified personnel operating under the applicable inspection standard.

XI. Defect Detection and Reduction

A. AI/ML in Defect Detection: Vision and Deep Learning

Deep learning (DL) has expanded the range of data-driven methods available for welding-defect detection. Convolutional neural networks (CNNs) learn image features jointly with the classifier, reducing—but not eliminating—manual decisions about preprocessing, labels, sampling, and validation. Reported performance therefore depends on the material, joint geometry, imaging system, and train-test separation used in each study [106].

TABLE IX
Common welding defects analyzed by LLM-based quality monitoring systems

Defect type	Semantic indicators	LLM-enabled analysis capability
Porosity	Gas entrapment, irregular void patterns, process instability	Semantic correlation with shielding gas flow, heat input, and process history
Lack of fusion	Incomplete bonding at joint interface, insufficient penetration	Reasoning over parameter settings and joint geometry to identify causal factors
Cracking	Linear discontinuities, thermal stress concentration	Cross-analysis of thermal cycles, material properties, and constraint conditions
Slag inclusion	Non-metallic inclusions trapped in weld metal	Context-aware interpretation of flux behavior and welding technique
Undercut	Groove formation along weld toe	Linking travel speed and heat input to geometric deviations

Different architectures address different inspection tasks. Faster R-CNN uses region proposals to localize candidate defects such as porosity or slag inclusions in radiographic images. Its suitability for nondestructive testing depends on defect prevalence, image resolution, calibration, and external validation rather than architecture alone [80].

Single-stage detectors such as YOLO (You Only Look Once) are often evaluated when latency matters. Welding studies have applied YOLO variants to surface-defect classes including cracks and burn-through, but laboratory frame rate does not by itself establish production deployment. YOLOv10-CASPD combines coordinate attention with space-to-depth convolution to target small structural features; Tables VI and VII retain the study-specific context for interpreting its results [64], [107].

For pixel-level analysis, U-Net variants are commonly used for semantic segmentation. They can delineate candidate defect regions such as lack of fusion or microcracks, although measurement uncertainty and acceptance decisions still require welding-specific validation [23].

B. LLMs for Defect Analysis: Root Cause Identification

Convolutional neural networks can localize or classify visible indications but do not by themselves establish why a defect formed. An LLM with retrieval may organize evidence from maintenance logs, inspection reports, and technical literature into candidate explanations. Those explanations remain hypotheses until checked against process data, metallurgical evidence, and expert review [79].

A notable example of quantitative defect modeling is demonstrated in work from The Pennsylvania State University, where the T2EGPT framework employs LLMs to derive physically interpretable equations governing humping defects in high-speed laser welding. Instead of

TABLE X
Performance metrics of AI/ML-based defect detection systems for welding

Metric	Definition	Relevance to weld inspection
Detection accuracy	Proportion of correctly identified defect and non-defect instances	Measures overall reliability of automated weld inspection systems [64]
Precision	Ratio of true positive detections to all predicted positives	Indicates robustness against false alarms in safety-critical weld inspection [80]
Recall (Sensitivity)	Ratio of true positive detections to all actual defects	Reflects the ability to detect rare or small defects such as micro-cracks and porosity [80]
Intersection over Union (IoU)	Overlap between predicted and ground-truth defect regions	Evaluates localization and segmentation quality for weld defects [23]
Inference latency	Time required to process a single image or frame	Critical for real-time robotic welding and in-process monitoring [64]
Robustness to noise	Stability of detection under imaging artifacts and process variability	Determines practical applicability in harsh industrial environments [106]

treating defect occurrence as a black-box classification problem, T2EGPT reveals that humping arises from an imbalance between inertia-dominated melt flow and capillary-driven surface stabilization. By expressing this relationship through dimensionless physical equations, the framework provides both predictive capability and mechanistic understanding, enabling informed process optimization [78].

In transferred aerospace-manufacturing evidence, CausalKGPT processes defect survey sheets and quality documentation through a causal quality-related knowledge graph. It offers a design pattern for structure-aware diagnosis, but its evaluation is not a welding experiment and does not by itself establish reliable weld-root-cause identification [16].

C. Automated Programming for Defect Detection Pipelines

Beyond analysis and reasoning, LLMs can assist with software pipelines for defect detection, including preprocessing, image mosaicking, and model configuration. A direct titanium-weld case study combines LLM-generated workflow assistance with a conventional ConvNeXt perception model; the code must still be reviewed, tested, and version-controlled [89].

In a typical bounded workflow, an LLM generates or modifies preprocessing code, while a separately trained and validated vision model performs defect localization or classification. Claims of reduced deployment time or cost require controlled measurements; the present evidence supports feasibility rather than a general industrial percentage improvement [89], [94].

TABLE XI
Qualitative energy-transfer characteristics of major welding processes; actual efficiency depends on equipment, joint, material, operating point, and system boundary

Welding process	Energy transfer mechanism	Energy efficiency characteristics
GTAW / GMAW	Diffuse plasma arc heat transfer to the weld pool	Low effective energy utilization due to radiation, spatter, and arc divergence
PAW / VPPAW	Constricted plasma arc with high energy density	Improved thermal efficiency and deeper penetration compared to conventional arc welding
Friction stir welding (FSW)	Direct mechanical-to-thermal energy conversion within the material	High energy efficiency due to solid-state joining and elimination of melting losses
Laser welding	Highly localized photon absorption in keyhole mode	Very high efficiency under stable keyhole conditions; sensitive to instability

XII. Energy Efficiency in Smart Welding Systems

A. Significance of Energy Efficiency in Welding Manufacturing

Welding concentrates energy into a joint while also consuming auxiliary power and materials. Heat input influences weld-pool behavior, microstructure, heat-affected zones, residual stress, and distortion, but energy and quality cannot be inferred from power alone [49], [50]. Defects can require repair, additional inspection, or rejection.

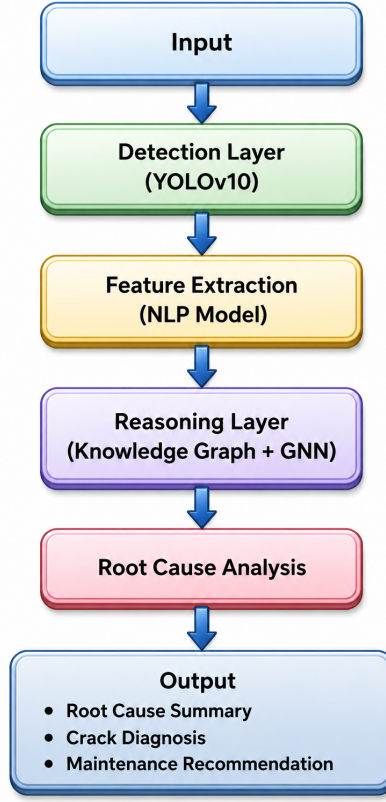


Fig. 10. Example of a generic LLM-based system architecture for defect detection and analysis

tion, thereby increasing the energy and material burden per accepted component.

At production scale, rework, scrap, idle equipment, fume extraction, shielding gas, and repeated inspection can materially change the system boundary [24]. From an Industry 5.0 perspective, energy efficiency should therefore be evaluated together with resilience, quality, and human well-being rather than as an isolated reduction in instantaneous power [45].

Excessive or poorly distributed heat input can alter weld-pool dynamics and solidification, contributing to coarsening, residual stress, distortion, or process instability [41], [50]. High-speed X-ray welding studies reveal key-hole fluctuations and pore-related mechanisms, but they do not by themselves quantify lifecycle energy penalties [51]. Energy consequences should therefore be measured rather than inferred from defect presence alone.

Conservative parameter selection can cause over-processing, whereas excessive energy reduction can cause lack of fusion. The relevant quantity is therefore energy per accepted joint under an approved procedure, not minimum heat input in isolation. Static welding-procedure specifications provide qualified operating envelopes; proposed adaptations outside those envelopes require requalification rather than automatic LLM execution [49].

A defensible energy study should report

$$E_{\text{accepted}} = \frac{E_{\text{process}} + E_{\text{idle}} + E_{\text{rework}} + E_{\text{inspection}}}{N_{\text{accepted}}}, \quad (1)$$

with material, joint geometry, quality class, production rate, metering boundary, and rejected-joint count stated explicitly. The present direct LLM corpus does not establish a general percentage reduction in welding energy.

B. Energy Characteristics of Major Welding Processes

Energy efficiency varies considerably across welding processes due to fundamental differences in heat generation mechanisms, energy transfer pathways, and modes of interaction between the heat source and base material. The proportion of supplied energy that effectively contributes to joint formation is governed by arc stability, heat source concentration, thermal conductivity, and process-specific losses arising from radiation, convection, and spatter. These mechanisms determine not only thermal efficiency but also the extent of HAZ development and residual stress accumulation [41], [49].

C. Interdependency Between Energy Input and Defect Formation

Energy input influences weld-pool dynamics, solidification, and thermal gradients. Depending on process and operating regime, excessive input can enlarge the

pool, increase distortion, or destabilize keyhole behavior, whereas insufficient input can limit penetration and fusion [41], [50], [51]. These relationships are process-specific and should not be reduced to a universal efficiency ranking.

Defect formation has a compounding effect on energy efficiency, as defective welds require repair welding, post-weld heat treatment, and additional inspection. These secondary operations significantly increase the lifecycle energy footprint, often outweighing energy savings achieved through initial parameter optimization. Energy-optimal welding therefore requires process stability rather than simplistic minimization of heat input. Stable energy delivery ensures consistent weld pool geometry, controlled solidification, and uniform microstructural evolution, enabling defect-free joints in a single pass.

D. Data-Driven and LLM-Enabled Energy Optimization

Traditional energy optimization strategies rely on static welding procedure specifications derived from empirical trials. While robust, these approaches are conservative and unable to adapt to real-time variations in material properties, joint fit-up, or environmental disturbances. As a result, excessive safety margins are commonly applied, leading to systematic over-processing and unnecessary energy consumption.

Data-driven optimization methods correlate parameters, quality indicators, and metered energy, but their transferability depends on data coverage. LLM-enabled systems may complement them by reasoning over procedure documents, temporally encoded sensor summaries, inspection reports, defect histories, and expert annotations; they do not replace metering, process models, or qualified control logic.

Crucially, LLM-enabled frameworks should function as decision-support systems rather than autonomous energy controllers. They can articulate trade-offs among measured energy, defect risk, throughput, and procedure constraints while preserving human authority. Any claimed savings require a controlled welding-specific experiment reporting the denominator above; no generic energy-saving percentage is claimed in this review.

XIII. Human-Centric Welding and Worker Interaction

A. Worker Safety and Training

Industry 5.0 explicitly prioritizes human well-being, sustainability, and resilience alongside productivity [6], [45]. Within this framing, AI and LLMs should augment skilled work and preserve human authority rather than being presented as replacements for qualified welders or inspectors.

A recent qualitative study of welding work shows that knowledge is distributed across embodied practice, documentation, peer interaction, and situated problem solving [108]. This evidence strengthens the rationale for grounded conversational support, multilingual explanation, and knowledge capture, while also showing why an

LLM interface cannot substitute for tacit expertise, local context, or accountable supervision.

1) **Enhancing Welder Safety through Automation of Hazardous Tasks:** Welding involves ultraviolet radiation, fumes, spatter, heat, electrical hazards, and demanding postures. Robotic systems can reduce exposure by taking repetitive or hazardous operations, while task allocation and residual risks still require formal assessment [18], [46].

Speed and Separation Monitoring and Power and Force Limiting are established collaborative-robot safety concepts. Sensor fusion may improve state estimation, but an AI prediction does not replace validated protective functions, risk assessment, or applicable safety standards [18]. Welding-specific human-safety outcomes remain an important evaluation gap.

2) **AI-Powered Training and AR/VR Guidance:** Extended Reality (XR), including Virtual and Augmented Reality, can support welding education and human-robot interaction. WELD-E provides a direct welding prototype combining XR, speech, VLM/LLM components, and ROS2, but its validation remains preliminary rather than evidence of generalized training transfer [48].

- **Immersive Training:** Trainees can rehearse procedures in a virtual environment, but benefits must be measured through retained skill, transfer to physical welding, and safety outcomes.
- **Performance Assessment:** Conventional sensing and ML can compare torch motion or process variables with expert demonstrations; this capability should not be attributed to an LLM unless the language model is actually evaluated.
- **In-situ AR Guidance:** AR overlays can present paths, safety zones, and instructions. Claims of reduced cognitive load or faster skill acquisition require controlled user studies [48].

B. LLM-Enabled Knowledge Sharing

LLMs can provide a natural-language interface to versioned welding procedures, standards, inspection history, and operator knowledge. Direct welding assistants now evaluate standards access and process Q&A, but the knowledge base—not the model’s parameters—must remain the authoritative, auditable record [90], [92].

1) **Natural Language Guidance and Troubleshooting:** LLM-based conversational systems can assist operators and engineers, subject to retrieval grounding, uncertainty disclosure, and escalation to qualified personnel [90], [92].

- **Interactive procedures and standards:** Welders can query controlled documents using natural language, while every answer preserves document version and clause-level provenance [90], [100].
- **Interpretative reasoning:** Knowledge-enhanced LLMs can generate root-cause hypotheses from retrieved welding evidence, but causal claims must be checked against process measurements and expert knowledge [16], [17], [35].

2) Bridging the Technical Barrier via Retrieval-Augmented Generation: RAG can ground outputs in validated technical documentation, inspection reports, and literature, but retrieval does not guarantee correctness: systems still require source display, contradiction checks, version control, abstention, and human review [82], [90], [92].

C. Ergonomic Assistance and Worker Well-Being

Ergonomics is a core component of human-centered manufacturing. Digital human modeling, biomechanical analysis, sensing, and exoskeletons provide relevant transferred evidence, but no included study demonstrates that an LLM itself reduces welding workload [45], [46].

1) Intelligent Assistance and Biometric Feedback:

- **Mental Fatigue Monitoring:** Physiological sensing may support workload estimation, but welding-specific accuracy, privacy, consent, drift, and intervention thresholds require dedicated validation.
- **Adaptive Robotic Assistance:** Workload-aware scheduling can adjust task allocation or pace, but safety constraints and human override remain primary [8], [18].
- **Exoskeleton Integration:** Industrial exoskeletons can reduce selected measures of physical load, but benefits depend on task and device and should not be generalized to LLM-enabled welding [46].

2) **Real-Time Ergonomic Risk Alerts:** As a prospective use case, deterministic posture models may compute indices such as RULA or REBA and pass a structured alert to an LLM for explanation. The scoring model, latency, privacy controls, and intervention effectiveness must be evaluated separately; no direct LLM-welding study in the corpus establishes prevention of long-term injury.

XIV. System Architecture

A. Architectural Design Principles

Proposed LLM-assisted welding platforms use layered cyber-physical architectures to separate sensing, deterministic control, analytical models, language-based reasoning, and human interaction. This separation supports the human-centered goals of Industry 5.0 while preventing the LLM from being treated as a real-time safety controller [45].

Key architectural objectives include:

- Seamless integration of physical welding equipment with cyber intelligence
- Real-time data acquisition and low-latency decision support
- Human-in-the-loop control with explainable AI outputs
- Scalability across welding processes and shop-floor configurations

These objectives are achieved through a multi-layered architecture that decouples sensing, communication, intelligence, and execution while maintaining closed-loop feedback for adaptive control.

B. Physical Layer

The physical layer constitutes the foundational interface between the welding environment and the cyber-intelligent system. It includes welding power sources (arc, laser, plasma), robotic manipulators, positioners, and auxiliary fixtures that directly interact with the workpiece. These components govern heat input, material flow, penetration depth, and solidification behavior. Advanced robotic welding cells provide high positional repeatability, while programmable power sources enable precise control of current, voltage, and waveform characteristics—parameters critical to weld quality and energy efficiency [1].

To capture underlying welding physics, the physical layer integrates multimodal sensing systems, including high-speed and RGB cameras, thermal and infrared sensors, acoustic emission sensors, and electrical signal monitoring units. Vision systems enable observation of weld pool geometry, spatter formation, and keyhole dynamics, while thermal sensors provide spatial temperature distributions for heat-affected zone (HAZ) assessment. Acoustic and electrical signals offer indirect yet highly sensitive indicators of arc instability, metal transfer modes, and defect initiation. Together, these sensors transform complex physical phenomena into digital data streams that feed higher-level reasoning and adaptive control modules [25].

C. Communication and Edge Layer

The communication and edge layer provides the real-time data transmission backbone connecting the physical layer to intelligent reasoning components. This layer employs Industrial Internet of Things (IIoT) protocols such as OPC UA and MQTT, supported by edge computing nodes deployed in proximity to the welding cell. Edge computing enables local signal preprocessing, feature extraction, and latency-critical inference, reducing reliance on cloud connectivity for safety- and quality-critical decisions [26].

Beyond latency reduction, the edge layer enhances system robustness by synchronizing multimodal sensor streams, filtering noise, and managing bandwidth efficiently. By performing localized inference tasks—such as arc stability detection or anomaly flagging—the system can continue bounded operation during network disturbances. This architectural choice aligns with Industry 5.0 objectives by balancing automation performance with resilience and reliability in harsh industrial environments [37].

D. Sensor-to-LLM Evidence Pipeline

Raw camera, acoustic, electrical, thermal, force, or photodiode streams must not be serialized directly into

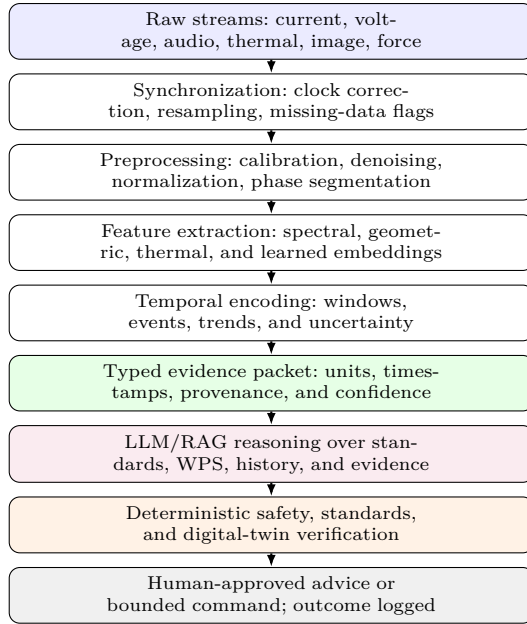


Fig. 11. Required sensor-to-LLM pipeline. Acquisition, synchronization, feature extraction, and hard real-time control remain outside the LLM.

a text prompt. First, every channel requires a shared time base, clock correction, resampling policy, calibration record, and explicit missingness flags. Second, signal-specific preprocessing removes noise, normalizes units, suppresses arc glare or transient artifacts, and segments process phases. Third, deterministic or learned perception models extract bead geometry, spectral bands, temperature trends, arc-stability indices, anomaly scores, image regions, and calibrated uncertainty. Fourth, temporal encoders aggregate these quantities into windows, events, trends, and candidate causal relationships. Only then is a typed evidence packet—with units, timestamps, provenance, uncertainty, and links to the underlying traces—provided to an LLM for retrieval, explanation, or planning.

E. Intelligence and Knowledge Layer

The intelligence and knowledge layer constitutes the cognitive core of the LLM-enabled smart welding architecture. Large Language Models (LLMs) within this layer interpret heterogeneous data sources, including sensor logs, welding procedure specifications, inspection reports, and historical defect databases. Unlike conventional machine learning models that focus on numerical optimization, LLMs enable semantic reasoning, contextual understanding, and explainable decision support [109].

Reliability is enhanced through Retrieval-Augmented Generation (RAG), which grounds LLM outputs in validated standards, experimental records, and domain-specific knowledge repositories [82]. Digital twins are integrated to test AI-generated recommendations through “what-if” analysis before physical execution [27]. The LLM therefore reasons over structured summaries and retrieved

evidence; it is not credited with the localization accuracy of a CNN, the temporal representation of a signal encoder, or the response time of an RL or PLC controller.

F. Human–Machine Interaction Layer

The human–machine interaction layer operationalizes the human-centric philosophy of Industry 5.0 by enabling intuitive collaboration between operators and intelligent systems. Natural language interfaces, augmented reality (AR), and visual dashboards translate complex system outputs into actionable guidance. AR overlays can highlight defect-prone regions or recommend real-time parameter adjustments, while LLM-based interfaces allow operators to query system behavior using conversational language [18].

This layer is intended to keep humans as active decision-makers rather than passive supervisors. Traceable explanations of defect hypotheses or energy trade-offs may support situational awareness and skill transfer, but improvements in trust, workload, or safety must be demonstrated with human-subject and operational evidence rather than assumed from the interface design [45].

G. Closed-Loop Feedback and Execution

Closed-loop feedback integrates all architectural layers, but free-form language is not placed directly in the hard real-time loop. Edge perception and conventional controllers analyze sensor data and execute certified control actions. An LLM may recommend a bounded parameter change or recovery plan only after schema, unit, procedure-envelope, digital-twin, interlock, and role-based authorization checks [22].

Each welding cycle may contribute versioned data to the knowledge base, but model or prompt updates must be validated before release rather than learned online without governance. Audit logs should preserve model/version, prompt template, retrieved evidence, proposed action, verifier outcomes, human decision, and executed command. This separation supports resilience and human authority while keeping the LLM outside the certified safety kernel.

XV. Case Studies and Applications

A. Academic Prototypes: Scientific Discovery and Framework Validation

Recent academic research has progressively moved beyond purely data-driven models toward physics-informed, language-reasoning, and digital-twin-integrated frameworks. These approaches test feasibility and interpretability but must be distinguished from routine industrial deployment [33], [78].

1) Pennsylvania State University (PSU): T2EGPT Framework: A Pennsylvania State University-led study introduced the Text-to-Equations Generative Pre-trained Transformer (T2EGPT) framework to predict humping onset in high-speed laser welding [78]. Humping is a

periodic surface instability that limits achievable welding speed and productivity.

- **Methodology:** T2EGPT combines LLM-derived relations from welding literature with sparse experimental data to generate an interpretable dimensionless equation linking melt-flow inertia and capillary stabilization.
- **Outcome:** The publisher record reports improved predictive accuracy and cross-material transferability relative to traditional data-driven models. No numerical value is reported here because the accessible primary record does not specify the page, table, metric definition, and denominator needed to validate the previously stated 95.83% result.

2) **Synchrotron X-ray-Based Monitoring:** Synchrotron X-ray studies provide direct observation of internal keyhole and melt-pool behavior that is difficult to infer from surface sensing. A verified copper laser-welding study used imaging at 20,000 frames/s to quantify time-dependent keyhole geometry [51].

- **Innovation:** High-speed X-ray imaging exposes internal geometry and dynamics rather than relying only on post-process sections or surface observations.
- **Application boundary:** Such measurements can support physics validation and future sensor-surrogate development, but the cited experiment does not itself demonstrate an LLM, CNN controller, or deployed low-cost sensor replacement.

3) **Zhejiang University: AI-Empowered Giga-Casting Design:** A multi-component AI-empowered framework has been reported for die design in giga-casting applications [44]. This is transferred manufacturing evidence, retained because multiphysics surrogates and constrained optimization offer a plausible design pattern for welding digital twins; it is not direct LLM-welding evidence.

- **Innovation:** The framework introduces multiphysical fields as intermediate variables, employing neural partial differential equation solvers to predict spatiotemporal distributions of fluid velocity and temperature.
- **Result:** By integrating deep reinforcement learning, the system emulates expert decision-making in optimizing gating systems and cooling channels, substantially reducing the cognitive load on engineers designing large-scale structural components.

B. Industrial Implementations: Intelligent Robotic Welding Cells

Industrial vendors offer robotic welding, automated optical inspection, seam tracking, and production-data platforms. These examples demonstrate conventional industrial automation and product capabilities; they are not counted as peer-reviewed LLM-welding experiments and should not be used to infer autonomous LLM control.

1) **Panasonic: Bead Eye Visual Inspection System:** Panasonic documents the Bead Eye vision-based inspection solution for selected arc-weld bead geometries [110].

- **Capability:** The vendor states that scanning, master comparison, and AI inspection can identify selected visible bead-shape anomalies and store inspection results for traceability.
- **Boundary:** The vendor also states that applicability must be demonstrated on the actual workpiece, internal defects and lack of penetration that are not visible in bead shape cannot be detected, and final workpiece quality remains the user’s responsibility [110]. The earlier “100% inspection coverage” claim was removed because no validated coverage denominator or detection guarantee was provided.

2) **Giga-Casting as Adjacent Manufacturing Context:** Giga-casting can consolidate assemblies and thereby change downstream joining requirements. The verified source retained here concerns an AI-empowered die-design framework, not a documented Tesla welding digital twin [44].

- **Transferable mechanism:** Multiphysics fields and learning-based optimization can assist design-space exploration for thermally coupled manufacturing processes.
- **Boundary:** Benefits for a particular vehicle platform, welding-joint count, residual stress, or energy consumption require product-specific primary evidence and are not claimed here.

3) **FANUC, YASKAWA, and Fincantieri:** Major industrial vendors provide vision and production-data platforms relevant to welding-intensive environments. The statements below describe vendor capabilities rather than controlled comparative experiments.

- **FANUC:** Integrated robot-vision functions can support workpiece localization and seam-related automation; implementation performance is application-specific.
- **YASKAWA:** YASKAWA Cockpit is documented as collecting and analyzing equipment data for condition monitoring and predictive-maintenance models [111].
- **Fincantieri:** The cited HRC source does not establish welding-specific digital-thread or energy outcomes. The example is therefore presented only as a prospective industrial use case, not as evidence of a deployed system.

C. Future Product Trends: LLMs, Digital Twins, and Large Reality Models

Future welding products may combine embodied intelligence, foundation models, and bounded adaptive control; this is a prospective direction rather than demonstrated deployment [19].

1) LLM-Powered Programming Interfaces: Natural-language interfaces may complement conventional programming for high-level task specification, while deterministic planners and robot controllers remain responsible for validated execution [112].

- Vision: Operators may issue commands such as “Prepare a fillet-weld plan for this T-joint.” An LLM may draft a task specification, but a typed planner, reachability and collision checks, process-window constraints, simulation, and human approval must precede robot execution.
- Potential advantage: If those safeguards are validated, the interface may reduce the skills barrier for configuring complex welding cells; the direct corpus does not yet establish the magnitude of this benefit.

2) Large Reality Models and Foundation Models: The integration of LLMs with digital twins motivates broader world-model or “large reality model” concepts; the terminology remains prospective and does not denote a certified welding product [113].

- Capability: LRMs enable natural-language-driven “what-if” simulations, where digital twins validate AI outputs before physical execution.
- Defect Detection: Visual foundation models and open-set recognition motivate investigation of unseen defect categories, but standalone welding acceptance requires calibrated welding-specific validation [64], [81].

D. Summary of Representative Studies and Applications as shown in Table IX

TABLE XII
Representative academic and industrial case studies on AI- and LLM-enabled welding systems

Category	Institution	Core Technology	Key Contribution
Academic	PSU	T2EGPT (LLM-based equation discovery)	Interpretable humping-onset equation from literature and sparse data; not a deployed controller
Academic	Univ. of Manchester	High-speed synchrotron X-ray	20,000-frame/s observation of copper keyhole dynamics; conventional physical evidence
Academic	Adjacent manufacturing	AI + multiphysics models	Transferred giga-casting die-design evidence; no welding or LLM validation
Industrial	Panasonic	Bead Eye visual inspection	Vendor-documented surface-bead inspection and traceability with explicit limitations
Industrial	Giga-casting context	AI-assisted die design	Adjacent design mechanism; product-specific joint, stress, and energy benefits not established here
Industrial	FANUC / YASKAWA	Robot vision + data platforms	Vendor capabilities; application-specific validation required
Future Trend	Multi-sector	LLMs + Digital Twins	Prospective natural-language planning behind deterministic verification and human approval

E. Source-Level Validation of Reported Experimental Values

Table XIII records the source-level checks for the manuscript’s most consequential numerical claims. Values are reported only in the task and validation context used by the cited study, together with a page, figure, table, or publisher-abstract locator. Untraceable values are identified as excluded rather than converted into qualitative evidence.

TABLE XIII
Audit trail for consequential quantitative claims

Source and evidence class	Validated value	Exact source locator	Interpretation used in this review
Günther et al. [73], C	Camera 1500 Hz at 144×176 pixels; three photodiodes at 40 kHz; 16 learned image features; reconstruction error 16.6% versus 15.5% for PCA	Sensor description, source p. 3; representation comparison, Table 2 on p. 7	Sensor and representation results; not an LLM experiment
Günther et al. [73], C	SVM classification error $0.81 \pm 0.11\%$ versus $1.79 \pm 0.19\%$ with PCA; nexting distance error 0.48 to 0.28; 82% four-class and 93.1% binary accuracy	Table 2, p. 7; Fig. 6 and discussion, pp. 10–13	Supports conventional learned prediction; the earlier 94% value was corrected
Günther et al. [73], C	30 simulator runs; 1.5–5 kW action range; 0.34 ms computation per control iteration	Controller experiment and discussion, pp. 12–14	Preliminary simulator control only; not a fully validated autonomous industrial cell
Lenka et al. [75], C	Tensile strength 142 ± 5.8 to 170 ± 4.2 MPa (+19.7%); IMC thickness 8.6 ± 0.7 to 6.6 ± 0.5 μm (−23.3%); conductivity +17.2%	Publisher abstract for article 114982	FSW experiment relative to the reported optimized static baseline; not LLM evidence
Gao et al. [76], C	55 rail-weld profiles; 5001 points/profile; convergence after 6000 training steps; 300 trade-off weights; > 220 feasible strategies/case; trough −0.41 to −0.18 mm	Publisher abstract and method description for article 113386	Rail-grinding optimization; 6000 denotes learning steps, not physical welding episodes
Tian et al. [15], T	64 samples; cross-validated R^2 : $89.85 \pm 6.17\%$ yield strength, $88.34 \pm 5.95\%$ tensile strength, $87.24 \pm 5.15\%$ elongation	Fig. 8 and accompanying text	Transferred steel-design evidence; not a welding trial
Lee et al. [78], E1	Qualitative validation of an interpretable humping-onset equation with cross-material transfer	Publisher record and article metadata, vol. 211, article 104320	A 95.83% value is excluded because an exact locator and metric definition were not available
Untraceable legacy claims	87% sensitivity, 92% specificity, 12% exploration-induced defect rate, 10^3 -fold sample reduction, 3000 training hours, 12,000 inspection records, and 100% inspection coverage	No supporting page, table, figure, or primary-source passage located	Excluded from the evidentiary synthesis

XVI. Limitations of the Review

The original historical database export was unavailable, so this revision does not reconstruct unverifiable historical counts. Its PRISMA numbers belong to a documented rerun in OpenAlex and Semantic Scholar, followed by backward checking and a Crossref/publisher-web top-up completed on 26 August 2026. This improves reproducibility and captured two recently indexed direct studies, but may still miss records indexed only in subscription databases, non-English sources, or papers released after the search date. General-web hit totals are unstable; accordingly, the top-up log reports the two additional unique eligible records actually retrieved and verified rather than inventing a search-engine result count. The direct corpus is also heterogeneous: image classification, knowledge retrieval, equation discovery, standards question answering, spatiotemporal prediction, robot planning, and conceptual demonstrations cannot be combined into a defensible pooled accuracy. Quality appraisal therefore remains domain-based, and the 21-study count should be interpreted as the result of the stated operational definition and search protocol rather than as a timeless census.

XVII. Conclusion and Future Outlook

This paper presents a systematic scoping review of LLM-enabled smart welding within the broader transition from Industry 4.0 to Industry 5.0. It retains the welding-science, conventional-AI, digital-twin, energy, and human-factors foundations needed to understand the field, while making the evidential boundary explicit. The reproducible update identified 21 direct welding publications: 15 quantitatively or comparatively evaluated studies, five prototypes or preliminary demonstrations, and one focused review. None establishes routine, safety-certified, continuously deployed closed-loop LLM control of an industrial welding process.

The evidence therefore supports a narrower but technically useful conclusion. Language-capable models can assist welding knowledge retrieval, standards access, multimodal inspection interpretation, equation discovery, process explanation, and high-level task planning. Conventional CNNs, temporal models, reinforcement learning, sensors, robot controllers, and digital twins remain the components that perform real-time perception, numerical estimation, and bounded execution. Adjacent-domain studies remain valuable when they demonstrate a transferable mechanism, but their reported performance cannot be relabeled as welding performance.

The welding analysis continues to show why integration matters. Heat input, weld-pool dynamics, geometry, metallurgy, defects, accepted production, and rework energy are coupled. An LLM may help operators retrieve procedures or interpret summarized evidence, yet high-frequency signals must first be synchronized, cleaned, transformed into physically meaningful features, temporally encoded, and accompanied by uncertainty and provenance. The LLM should remain outside the deterministic safety kernel; digital-twin checks, standards constraints, conventional controllers, and human authorization must gate any proposed action.

Human-centered use is consequently more credible than claims of worker replacement or unsupervised autonomy. Near-term applications include cited answers over approved procedures, explanation of alarms, training support, structured incident reports, and assisted programming. Claims that these tools reduce workload, energy, defects, or skills shortages require task-specific controlled studies with welder participation, clearly defined baselines, uncertainty analysis, and measurement of both productivity and safety outcomes.

Priority research should include shared welding-specific benchmarks; temporal multimodal datasets with traceable sensor calibration; prospective comparisons against expert and conventional-AI baselines; ablations of retrieval, vision, and language components; hallucination and out-of-distribution tests; latency and fail-safe evaluation; and multi-site industrial studies. Energy claims should report system boundaries and accepted-part energy rather than model inference alone. Robot studies should distinguish language interpretation from verified planning and certified motion control.

LLMs thus represent an emerging knowledge and interaction layer for smart welding, not yet a demonstrated autonomous control layer. The strongest path forward is a hybrid, auditable architecture that combines welding physics and metallurgy, validated perception and control, digital twins, curated retrieval, uncertainty-aware language interfaces, and final human authority. This evidence-calibrated framing preserves the field's ambition while providing a defensible basis for future Industry 5.0 welding research.

Declarations

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Data Availability: Search records, screening decisions, direct-evidence classifications, quality appraisal, DOI audit, and the reference-change register are supplied as

machine-readable supplementary files. No new experimental welding dataset was generated.

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